Grade 2 Mathematics

Balochistan Textbook Board

Solved Exercises — Complete Book

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Unit 1 — Whole Numbers (pages 5 to 29)

Students' learning outcomes (page 5): After completing this unit, students will be able to: count numbers up to and across 999 (3 digit numbers) forward and backward, beginning from zero or one or from any given number; read and write numbers up to 999 in numeral and up to 99 in words; recognize the place value of each digit in 3-digit numbers (hundred, tens, ones / unit); compare and order number up to 999 using appropriate language (for instance: more and less, greater, smaller, equal to, same as, increasing, decreasing, smallest to largest and vice versa etc.); round numbers to the nearest tens using different concrete objects and pictorial representation; recognize the position of objects and write it using ordinal numbers up to 20; read and write Roman numbers up to 12.

Numbers in Words up to 99 (pages 6-7) — Read and write numbers up to 99

Number — word (pages 6-7)

1 — One

2 — Two

3 — Three

4 — Four

5 — Five

6 — Six

7 — Seven

8 — Eight

9 — Nine

10 — Ten

11 — Eleven

12 — Twelve

13 — Thirteen

14 — Fourteen

15 — Fifteen

16 — Sixteen

17 — Seventeen

18 — Eighteen

19 — Nineteen

20 — Twenty

21 — Twenty one

22 — Twenty two

23 — Twenty three

24 — Twenty four

25 — Twenty five

26 — Twenty six

27 — Twenty seven

28 — Twenty eight

29 — Twenty nine

30 — Thirty

31 — Thirty one

32 — Thirty two

33 — Thirty three

34 — Thirty four

35 — Thirty five

36 — Thirty six

37 — Thirty seven

38 — Thirty eight

39 — Thirty nine

40 — Forty

41 — Forty one

42 — Forty two

43 — Forty three

44 — Forty four

45 — Forty five

46 — Forty six

47 — Forty seven

48 — Forty eight

49 — Forty nine

50 — Fifty

51 — Fifty one

52 — Fifty two

53 — Fifty three

54 — Fifty four

55 — Fifty five

56 — Fifty six

57 — Fifty seven

58 — Fifty eight

59 — Fifty nine

60 — Sixty

61 — Sixty one

62 — Sixty two

63 — Sixty three

64 — Sixty four

65 — Sixty five

66 — Sixty six

67 — Sixty seven

68 — Sixty eight

69 — Sixty nine

70 — Seventy

71 — Seventy one

72 — Seventy two

73 — Seventy three

74 — Seventy four

75 — Seventy five

76 — Seventy six

77 — Seventy seven

78 — Seventy eight

79 — Seventy nine

80 — Eighty

81 — Eighty one

82 — Eighty two

83 — Eighty three

84 — Eighty four

85 — Eighty five

86 — Eighty six

87 — Eighty seven

88 — Eighty eight

89 — Eighty nine

90 — Ninety

91 — Ninety one

92 — Ninety two

93 — Ninety three

94 — Ninety four

95 — Ninety five

96 — Ninety six

97 — Ninety seven

98 — Ninety eight

99 — Ninety nine

1-999 Number Chart (page 8) — forward

The chart counts in tens: 1 to 9, then 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, and so on up to 990, 999. Reading across a row adds 10 each time; the last row is 910, 920, 930, 940, 950, 960, 970, 980, 990, 999.

999-1 Number Chart (page 9) — backward

The chart counts backwards in tens: 9 down to 1, then 999, 990, 980, 970, 960, 950, 940, 930, 920, 910, and so on down to 100, 90, 80, 70, 60, 50, 40, 30, 20, 10.

Roman Numeral up to 12 (page 10)

Number — Roman numeral

1 — I

2 — II

3 — III

4 — IV

5 — V

6 — VI

7 — VII

8 — VIII

9 — IX

10 — X

11 — XI

12 — XII

Remember (page 10): Usual numbers 1, 5, 10 are I, V, X. To make 4 and 9 we subtract: V - I = IV = 4; X - I = IX = 9. For 6, 7, 8, 11 and 12 we add: V + I = VI = 6; V + II = VII = 7; V + III = VIII = 8; X + I = XI = 11; X + II = XII = 12.

Read the Roman numerals (page 11)

Q: III is called?

A: 3.

Q: VI is called?

A: 6. NOTE — BOOK ERROR: the book prints 'IV is called 6' here; IV is 4 and VI is 6, so the intended numeral is VI.

Q: IX is called?

A: 9.

Q: XII is called?

A: 12.

Write the time by looking at the clocks (page 11)

Q: First clock (short hand at 3, long hand at 12):

A: 3 o'clock.

Q: Second clock (short hand at 6, long hand at 12):

A: 6 o'clock.

Q: Third clock (short hand at 9, long hand at 12):

A: 9 o'clock.

Write the missing Roman numerals (page 11) — usual number for each Roman numeral

Roman numeral — usual number

III — 3

VI — 6

IX — 9

XI — 11

I — 1

IV — 4

Ordinal Numbers (page 12)

Position — ordinal

1st — First

2nd — Second

3rd — Third

4th — Fourth

5th — Fifth

6th — Sixth

7th — Seventh

8th — Eighth

9th — Ninth

10th — Tenth

11th — Eleventh

12th — Twelfth

13th — Thirteenth

14th — Fourteenth

15th — Fifteenth

16th — Sixteenth

17th — Seventeenth

18th — Eighteenth

19th — Nineteenth

20th — Twentieth

Exercise 1.1 — 1. Match the columns (page 13)

Number — word

19 — Nineteen

28 — Twenty eight

37 — Thirty seven

46 — Forty six

55 — Fifty five

64 — Sixty four

75 — Seventy five

84 — Eighty four

93 — Ninety three

100 — Hundred

Exercise 1.1 — 2. Write the numbers in words (page 14)

Number — word

34 — Thirty four

27 — Twenty seven

63 — Sixty three

62 — Sixty two

88 — Eighty eight

12 — Twelve

48 — Forty eight

59 — Fifty nine

77 — Seventy seven

95 — Ninety five

Exercise 1.1 — 3. Write the position of the shapes (page 14)

Following the arrows from the first shape: 1st square, 2nd circle, 3rd oval, 4th triangle, 5th rectangle, 6th square, 7th triangle, 8th circle, 9th rectangle, 10th oval, 11th circle, 12th triangle, 13th square, 14th rectangle, 15th oval, 16th square, 17th triangle, 18th rectangle, 19th circle, 20th oval.

Exercise 1.1 — 4. Write missing Roman numerals (page 15)

Usual number — Roman numeral

1 — I (given)

2 — II

3 — III

4 — IV

5 — V (given, printed as 'v')

6 — VI

7 — VII (given, printed as 'vii')

8 — VIII

9 — IX

10 — X (given, printed as 'x')

3-Digit Numbers (page 15)

We have already learnt numbers up to 100. Counting in tens: 10 ones make 1 ten; 2 tens make 20; 3 tens make 30. Key point: 10 tens = 100 (one hundred). 100 is the smallest 3-digit number. Now let us count in hundreds with the help of blocks.

Counting in tens (page 16)

Tens — number

4-Tens — 40

5-Tens — 50

6-Tens — 60

7-Tens — 70

8-Tens — 80

9-Tens — 90

10-Tens — 100 = 1 hundred

Counting in hundreds (page 17)

Hundreds — number

1 hundred — 100

2 hundred — 200

3 hundred — 300

4 hundred — 400

5 hundred — 500

6 hundred — 600

7 hundred — 700

8 hundred — 800

9 hundred — 900

Activity 1 (page 18) — place value with digit cards 4, 2, 5

The three students hold 4, 2, 5, making the number 425. 4 is at hundreds place, so its place value is 4 x 100 = 400. 2 is at tens place, so its place value is 2 x 10 = 20. 5 is at ones place, so its place value is 5 x 1 = 5. The number is 400 + 20 + 5 = 425. Key point: 10 = 1 ten = 10 ones; 100 = 1 hundred = 10 tens. Changing the students' positions makes a new number (class activity).

Place Value of 3-digit Number (page 19)

The position of a digit in a number is called its place value. Each place has a specific name and value. 100 is the first 3-digit number: H=1, T=0, O=0. The place value of 101: H=1, T=0, O=1. For 137: 1 hundred block, 3 ten rods, 7 ones cubes — H=1, T=3, O=7. The digit 1 is at the hundred place, so its value is 100. The digit 3 is at the tens place, so its value is 30. The digit 7 is at the ones place, so its value is 7.

Place value of 789 and 999 (page 20)

789: H=7, T=8, O=9. 7 is at hundred's place, so its value is 700. 8 is at ten's place, so its value is 80. 9 is at one's place, so its value is 9. 999: H=900, T=90, O=9.

Exercise 1.2 — 1. Write the missing number (page 20)

Row — complete sequence

500, 501, _, _, 504, _, 506 — 500, 501, 502, 503, 504, 505, 506

507, 508, _, _, _, 512, _ — 507, 508, 509, 510, 511, 512, 513

514, 515, _, _, 518, _, _ — 514, 515, 516, 517, 518, 519, 520

521, _, _, 524, _, _, 527 — 521, 522, 523, 524, 525, 526, 527

528, _, _, _, 532, _, _ — 528, 529, 530, 531, 532, 533, 534

Exercise 1.2 — 2. Write the place value of the digit in the number (page 21)

Q: (i) 434:

A: Hundreds: 4 = 400; Tens: 3 = 30; Ones: 4 = 4.

Q: (ii) 567:

A: Hundreds: 5 = 500; Tens: 6 = 60; Ones: 7 = 7.

Q: (iii) Blocks shown (1 hundred flat, 4 ten rods, 7 ones cubes):

A: The number is 147. Hundreds: 1 = 100; Tens: 4 = 40; Ones: 7 = 7.

Exercise 1.2 — 3. Write the place value of the circled digit (page 21)

Number (circled digit) — place value

5(6)3 — 60 (given in the book)

38(7) — 7

(4)68 — 400

(2)75 — 200

Exercise 1.2 — 4. Write the place value of given numbers (page 21)

Number — Ones, Tens, Hundreds

639 — 9 Ones, 3 Tens, 6 Hundreds (given in the book)

360 — 0 Ones, 6 Tens, 3 Hundreds

968 — 8 Ones, 6 Tens, 9 Hundreds

512 — 2 Ones, 1 Ten, 5 Hundreds

309 — 9 Ones, 0 Tens, 3 Hundreds

758 — 8 Ones, 5 Tens, 7 Hundreds

Comparing and Ordering Numbers (pages 22-23)

Symbols: '<' means 'less than', '>' means 'greater than', '=' means 'equal to'. NOTE — BOOK ERROR on page 22: the '<' example is printed as '75 < 70', which is false; the correct example is 70 < 75 (the '>' example, 75 > 70, is correct). Key point: the best way of comparing numbers is to compare each number's place value starting from the leftmost digit. Step 1: Compare 728 and 396. Hundreds digits 7 and 3: 7 is greater than 3, so 728 > 396. Step 2: If the hundreds digits are the same, compare the tens digits: 728 and 796 both have 7 hundreds; tens 2 is less than 9, so 728 < 796. Step 3: If hundreds and tens are the same, compare the ones digits: 798 and 796 both have 7 hundreds and 9 tens; ones 8 is greater than 6, so 798 > 796. Step 4: If all digits are the same, the numbers are

equal: 467 = 467. Key point: a 3-digit number is always greater than a 2-digit number.

Smaller to larger (page 23): 20, 10, 15 in smaller-to-larger order is 10, 15, 20 (1 ten is smallest, 2 tens is greatest).

Ordering examples (page 24)

351, 162 and 479 from small to large: compare hundreds digits 3, 1, 4 — 1 is smallest and 4 is largest, so the order is 162 < 351 < 479. 382, 416 and 235 from large to small: hundreds digits 3, 4, 2 — 4 is greater than 3 and 3 is greater than 2, so the order is 416 > 382 > 235.

Exercise 1.3 — 1. Encircle the smallest number (page 25)

Pair — smallest

(i) 201, 405 — 201

(ii) 334, 601 — 334

(iii) 701, 890 — 701

(iv) 241, 280 — 241

(v) 99, 80 — 80

(vi) 90, 80 — 80

Exercise 1.3 — 2. Encircle the larger number (page 25)

Pair — larger

(i) 412, 501 — 501

(ii) 637, 688 — 688

(iii) 560, 566 — 566

(iv) 893, 712 — 893

(v) 75, 79 — 79

(vi) 737, 748 — 748

Exercise 1.3 — 3. Write the numbers from smaller to larger (page 25)

Set — smaller to larger

(i) 40, 70, 30 — 30, 40, 70

(ii) 107, 180, 115 — 107, 115, 180

(iii) 40, 72, 35 — 35, 40, 72

(vi) 880, 850, 890 — 850, 880, 890

Note: item (iv) is skipped in the book's numbering on this page — the printed sets are (i), (ii), (iii) and (vi).

Exercise 1.3 — 4. Write the numbers from larger to smaller (page 25)

Set — larger to smaller

(i) 40, 28, 80 — 80, 40, 28

(ii) 825, 225, 525 — 825, 525, 225

(iii) 35, 75, 67 — 75, 67, 35

(iv) 550, 650, 399 — 650, 550, 399

Exercise 1.3 — 5. Put the signs <, > and = (page 25)

Pair — sign

(i) 87 ___ 78 — 87 > 78

(ii) 56 ___ 54 — 56 > 54

(iii) 197 ___ 211 — 197 < 211

(iv) 408 ___ 470 — 408 < 470

(v) 665 ___ 665 — 665 = 665

(vi) 501 ___ 501 — 501 = 501

Round Numbers to the nearest tens (page 26)

Rounding to the nearest ten means approximating a number to the closest multiple of 10. Rules: look at the ones digit (the rightmost digit). If the ones digit is 0, 1, 2, 3 or 4, round down to the nearest ten. If the ones digit is 5, 6, 7, 8 or 9, round up to the nearest ten. Examples: 8 rounds up to 10; 18 rounds up to 20.

Example 1: Rounding 46 to the nearest ten — the ones digit is 6; since 6 is 5 or more, round up; the nearest ten is 50. Example 2: Rounding 89 to the nearest ten — the nearest ten is 90. NOTE — BOOK ERROR: the book's reasoning for Example 2 is printed as 'Since 9 is less than 5, round down', which contradicts its own rule and its own answer; the correct reasoning is that 9 is 5 or more, so 89 rounds UP to 90.

Let us round off the numbers to the nearest tens (page 27)

Number — nearest ten

28 — 30

8 — 10

28 — 30

15 — 20

Exercise 1.4 — 1. Round off the following numbers to the nearest tens (page 27)

Number — nearest ten

18 — 20

34 — 30

168 — 170

368 — 370

29 — 30

41 — 40

252 — 250

425 — 430

Exercise 1.4 — 2. Tick the nearest tens (page 27)

17 rounds to the nearest ten:

- 10
- ✓ 20
- 30

82 rounds to the nearest ten:

- ✓ 80
- 70
- 90

I have Learnt (page 28)

Counting numbers up to 3-digit number (backward and forward); reading and writing numbers up to 3-digit numbers in numerals; reading and writing number up to 99 in words; recognizing the place value of each digit in 3-digit numbers (Hundreds, Tens, ones); comparing and ordering numbers up to 999 using <, > and = signs; rounding off the numbers to the nearest tens; recognizing the position of objects using ordinal

numbers up to 20; reading and writing Roman numbers up to 12.

Vocabulary used: Whole numbers, place value, ordinal numbers, Roman numbers, position, compare, order, hundreds, tens, ones, units, round off, nearest.

Review Exercise — 1. Tick the correct answer (page 28)

(i) The place value of 7 in the number 875 is:

- Ones
- ✓ Tens
- Hundreds

(ii) To round off 28 to the nearest tens:

- ✓ 30
- 25
- 20

(iii) To round off 93 to the nearest tens:

- 100
- ✓ 90
- 80

(iv) V + I =

- ✓ VI
- VII
- IV

(v) Larger to the smaller arrangement of the numbers 879, 789, 978 is:

- ✓ 978, 879, 789
- 879, 978, 789
- 978, 789, 879

Review Exercise — 2. Write in words the following numbers (page 29)

Number — word

(i) 17 — Seventeen

(ii) 37 — Thirty seven

(ii, printed again — third item) 86 — Eighty six

(iv) 56 — Fifty six

(v) 20 — Twenty

NOTE: the book numbers the 86 item as '(ii)' a second time; it is the third item in the list.

Review Exercise — 3. Write the place value of the circled digit (page 29)

Number (circled digit) — place value

(i) 4(3)7 — 30

(ii) (8)05 — 800

(iii) 97(2) — 2

(iv) (6)79 — 600

Review Exercise — 4. Arrange the numbers (page 29)

Q: 412, 439, 399, 385, 499 — Ascending order (smaller to larger):

A: 385, 399, 412, 439, 499.

Q: 412, 439, 399, 385, 499 — Descending order (larger to smaller):

A: 499, 439, 412, 399, 385.

Review Exercise — 5. Write the Roman numerals of the following

(page 29)

Number — Roman numeral

4 — IV

5 — V

6 — VI

8 — VIII

10 — X

12 — XII

Review Exercise — 6. Use the correct signs <, > and = (page 29)

Pair — sign

(i) 28 ___ 23 — 28 > 23

(ii) 350 ___ 289 — 350 > 289

(iii) 509 ___ 509 — 509 = 509

(iv) 789 ___ 789 — 789 = 789

(v) 812 ___ 788 — 812 > 788

(vi) 987 ___ 879 — 987 > 879

Unit 2 — Number Operation: Addition and Subtraction (pages 30 to 51)

Students' learning outcomes (page 30): After completing this unit, students will be able to: find, recall and use addition and subtraction facts to 100; add and subtract numbers mentally and in formal written form

(with and without regrouping) including 3-digit number and 1-digit number, 3-digit number and tens, 3-digit number and a 2-digit number, two 3-digit numbers; solve real-world problems with addition and subtraction (involving missing numbers, money, quantities and measures); estimate the answer to an addition and subtraction question (using various approaches).

Unit opener leaf sums (page 30)

35 + 7

Step 1: Start at 35.

Step 2: Add 7: 35 + 5 = 40, then 40 + 2 = 42.

Answer: 42 (given in the book)

6 + 24

Step 1: Start at 24.

Step 2: Add 6: 24 + 6 = 30.

Answer: 30

17 + 8

Step 1: Start at 17.

Step 2: Add 8: 17 + 3 = 20, then 20 + 5 = 25.

Answer: 25

Recall addition facts to 100 (page 31)

Addition of two 1-digit numbers: 5 mangoes + 3 mangoes = 8 mangoes, so 5 + 3 = 8. Addition of a 2-digit number and a 1-digit number: 12 + 4 — Step 1, add ones: 2 ones + 4 ones = 6 ones, write 6 in the Ones column; Step 2, add tens: 1 ten + 0 tens = 1 ten, write 1 in the Tens

column; answer 16. Addition of two 2-digit numbers: 35 + 43 — Step 1, add ones: 5 ones + 3 ones = 8 ones; Step 2, add tens: 3 tens + 4 tens = 7 tens; answer 78.

Exercise 2.1 — Add the following (page 32)

(1) 37 + 2

Step 1: Add ones: 7 ones + 2 ones = 9 ones.

Step 2: Add tens: 3 tens + 0 tens = 3 tens.

Answer: 39

(2) 72 + 5

Step 1: Add ones: 2 ones + 5 ones = 7 ones.

Step 2: Add tens: 7 tens + 0 tens = 7 tens.

Answer: 77

(3) 63 + 5

Step 1: Add ones: 3 ones + 5 ones = 8 ones.

Step 2: Add tens: 6 tens + 0 tens = 6 tens.

Answer: 68

(4) 83 + 4

Step 1: Add ones: 3 ones + 4 ones = 7 ones.

Step 2: Add tens: 8 tens + 0 tens = 8 tens.

Answer: 87

(5) 25 + 12

Step 1: Add ones: 5 ones + 2 ones = 7 ones.

Step 2: Add tens: 2 tens + 1 ten = 3 tens.

Answer: 37

(6) 32 + 22

Step 1: Add ones: 2 ones + 2 ones = 4 ones.

Step 2: Add tens: 3 tens + 2 tens = 5 tens.

Answer: 54

(7) 43 + 45

Step 1: Add ones: 3 ones + 5 ones = 8 ones.

Step 2: Add tens: 4 tens + 4 tens = 8 tens.

Answer: 88

(8) 56 + 42

Step 1: Add ones: 6 ones + 2 ones = 8 ones.

Step 2: Add tens: 5 tens + 4 tens = 9 tens.

Answer: 98

Addition of 3-digit and 1-digit numbers, without regrouping (page 32)

342 + 6 — Step 1, add ones: 2 ones + 6 ones = 8 ones, write 8 in Ones place; Step 2, add tens: 4 tens + 0 tens = 4 tens, write 4 in Tens place; Step 3, add hundreds: 3 hundreds + 0 hundreds = 3 hundreds, write 3 in Hundreds place; answer 348.

Addition of 3-digit and 1-digit numbers, with regrouping (page 33)

178 + 6 — Step 1, add ones: 8 ones + 6 ones = 14 ones = 1 ten + 4 ones, write 4 in Ones place; Step 2, carry 1 ten to the tens place, then add tens: 1 ten + 7 tens = 8 tens, write 8 in Tens place; Step 3, add hundreds: 1 hundred + 0 hundreds = 1 hundred, write 1 in Hundreds

place; answer 184.

Addition of 3-digit numbers and Tens (page 33)

254 + 20 — Step 1, add ones: 4 ones + 0 ones = 4 ones; Step 2, add tens: 5 tens + 2 tens = 7 tens; Step 3, add hundreds: 2 hundreds + 0 hundreds = 2 hundreds; answer 274.

Exercise 2.2 — Add the following (pages 33-34)

(1) 496 + 7

Step 1: Add ones: 6 ones + 7 ones = 13 ones = 1 ten + 3 ones; write 3, carry 1 ten.

Step 2: Add tens: 1 ten + 9 tens = 10 tens = 1 hundred + 0 tens; write 0, carry 1 hundred.

Step 3: Add hundreds: 1 hundred + 4 hundreds = 5 hundreds.

Answer: 503

(2) 968 + 4

Step 1: Add ones: 8 ones + 4 ones = 12 ones = 1 ten + 2 ones; write 2, carry 1 ten.

Step 2: Add tens: 1 ten + 6 tens = 7 tens.

Step 3: Add hundreds: 9 hundreds + 0 hundreds = 9 hundreds.

Answer: 972

(3) 682 + 2

Step 1: Add ones: 2 ones + 2 ones = 4 ones.

Step 2: Add tens: 8 tens + 0 tens = 8 tens.

Step 3: Add hundreds: 6 hundreds + 0 hundreds = 6 hundreds.

Answer: 684

(4) 135 + 2

Step 1: Add ones: 5 ones + 2 ones = 7 ones.

Step 2: Add tens: 3 tens + 0 tens = 3 tens.

Step 3: Add hundreds: 1 hundred + 0 hundreds = 1 hundred.

Answer: 137

(5) 456 + 3

Step 1: Add ones: 6 ones + 3 ones = 9 ones.

Step 2: Add tens: 5 tens + 0 tens = 5 tens.

Step 3: Add hundreds: 4 hundreds + 0 hundreds = 4 hundreds.

Answer: 459

(6) 329 + 3

Step 1: Add ones: 9 ones + 3 ones = 12 ones = 1 ten + 2 ones; write 2, carry 1 ten.

Step 2: Add tens: 1 ten + 2 tens = 3 tens.

Step 3: Add hundreds: 3 hundreds + 0 hundreds = 3 hundreds.

Answer: 332

Addition of 3-digit and 2-digit numbers, without regrouping (page 34)

154 + 25 — Step 1, add ones: 4 ones + 5 ones = 9 ones, write 9 in Ones place; Step 2, add tens: 5 tens + 2 tens = 7 tens, write 7 in Tens place; answer 179. NOTE — BOOK TYPO: Step 2 is printed as 'Write 4 in tens place'; the correct tens digit is 7, as in the book's own solution box.

Addition of 3-digit and 2-digit numbers, with regrouping (page

34)

347 + 36 — Step 1, add ones: 7 ones + 6 ones = 13 ones = 1 ten + 3 ones, write 3 in Ones place; Step 2, carry 1 ten to the tens place, then add tens: 1 ten + 4 tens + 3 tens = 8 tens, write 8 in Tens place; Step 3, add hundreds: 3 hundreds + 0 hundreds = 3 hundreds; answer 383.

Exercise 2.3 — Add the following (page 35)

(1) 135 + 32

Step 1: Add ones: 5 ones + 2 ones = 7 ones.

Step 2: Add tens: 3 tens + 3 tens = 6 tens.

Step 3: Add hundreds: 1 hundred + 0 hundreds = 1 hundred.

Answer: 167

(2) 456 + 43

Step 1: Add ones: 6 ones + 3 ones = 9 ones.

Step 2: Add tens: 5 tens + 4 tens = 9 tens.

Step 3: Add hundreds: 4 hundreds + 0 hundreds = 4 hundreds.

Answer: 499

(3) 328 + 61

Step 1: Add ones: 8 ones + 1 one = 9 ones.

Step 2: Add tens: 2 tens + 6 tens = 8 tens.

Step 3: Add hundreds: 3 hundreds + 0 hundreds = 3 hundreds.

Answer: 389

(4) 124 + 38

Step 1: Add ones: 4 ones + 8 ones = 12 ones = 1 ten + 2 ones; write 2, carry 1 ten.

Step 2: Add tens: 1 ten + 2 tens + 3 tens = 6 tens.

Step 3: Add hundreds: 1 hundred + 0 hundreds = 1 hundred.

Answer: 162

(5) 569 + 26

Step 1: Add ones: 9 ones + 6 ones = 15 ones = 1 ten + 5 ones; write 5, carry 1 ten.

Step 2: Add tens: 1 ten + 6 tens + 2 tens = 9 tens.

Step 3: Add hundreds: 5 hundreds + 0 hundreds = 5 hundreds.

Answer: 595

(6) 247 + 14

Step 1: Add ones: 7 ones + 4 ones = 11 ones = 1 ten + 1 one; write 1, carry 1 ten.

Step 2: Add tens: 1 ten + 4 tens + 1 ten = 6 tens.

Step 3: Add hundreds: 2 hundreds + 0 hundreds = 2 hundreds.

Answer: 261

Addition of two 3-digit numbers, without regrouping (page 35)

453 + 245 — Step 1, add ones: 3 ones + 5 ones = 8 ones, write 8 in Ones place; Step 2, add tens: 5 tens + 4 tens = 9 tens, write 9 in Tens place; Step 3, add hundreds: 4 hundreds + 2 hundreds = 6 hundreds, write 6 in Hundreds place; answer 698.

Addition of two 3-digit numbers, with regrouping (page 36)

365 + 276 — Step 1, add ones: 5 ones + 6 ones = 11 ones = 1 ten + 1 one, write 1 in Ones place; Step 2, carry 1 ten to the tens place, then add tens: 1 ten + 6 tens + 7 tens = 14 tens = 1 hundred + 4 tens, write 4 in

Tens place; Step 3, carry 1 hundred to the Hundreds place, then add hundreds: 1 hundred + 3 hundreds + 2 hundreds = 6 hundreds, write 6 in Hundreds place; answer 641.

Exercise 2.4 — Add the following (page 36)

(1) 433 + 335

Step 1: Add ones: 3 ones + 5 ones = 8 ones.

Step 2: Add tens: 3 tens + 3 tens = 6 tens.

Step 3: Add hundreds: 4 hundreds + 3 hundreds = 7 hundreds.

Answer: 768

(2) 315 + 443

Step 1: Add ones: 5 ones + 3 ones = 8 ones.

Step 2: Add tens: 1 ten + 4 tens = 5 tens.

Step 3: Add hundreds: 3 hundreds + 4 hundreds = 7 hundreds.

Answer: 758

(3) 123 + 334

Step 1: Add ones: 3 ones + 4 ones = 7 ones.

Step 2: Add tens: 2 tens + 3 tens = 5 tens.

Step 3: Add hundreds: 1 hundred + 3 hundreds = 4 hundreds.

Answer: 457

(4) 376 + 236

Step 1: Add ones: 6 ones + 6 ones = 12 ones = 1 ten + 2 ones; write 2, carry 1 ten.

Step 2: Add tens: 1 ten + 7 tens + 3 tens = 11 tens = 1 hundred + 1 ten; write 1, carry 1 hundred.

Step 3: Add hundreds: 1 hundred + 3 hundreds + 2 hundreds = 6 hundreds.

Answer: 612

(5) 148 + 263

Step 1: Add ones: 8 ones + 3 ones = 11 ones = 1 ten + 1 one; write 1, carry 1 ten.

Step 2: Add tens: 1 ten + 4 tens + 6 tens = 11 tens = 1 hundred + 1 ten; write 1, carry 1 hundred.

Step 3: Add hundreds: 1 hundred + 1 hundred + 2 hundreds = 4 hundreds.

Answer: 411

(6) 254 + 267

Step 1: Add ones: 4 ones + 7 ones = 11 ones = 1 ten + 1 one; write 1, carry 1 ten.

Step 2: Add tens: 1 ten + 5 tens + 6 tens = 12 tens = 1 hundred + 2 tens; write 2, carry 1 hundred.

Step 3: Add hundreds: 1 hundred + 2 hundreds + 2 hundreds = 5 hundreds.

Answer: 521

Real-world problem involving addition (page 37)

(i) There are 38 students in class-1 and 27 students in class-2. How many students are there in all?

Step 1: Add ones: 8 ones + 7 ones = 15 ones = 1 ten + 5 ones; write 5, carry 1 ten.

Step 2: Add tens: 1 ten + 3 tens + 2 tens = 6 tens.

Answer: 65 students in all

(ii) Amjad has Rs. 258. His father gave him Rs. 35 more. How much money has he now?

Step 1: Add ones: 8 ones + 5 ones = 13 ones = 1 ten + 3 ones; write 3, carry 1 ten.

Step 2: Add tens: 1 ten + 5 tens + 3 tens = 9 tens.

Step 3: Add hundreds: 2 hundreds + 0 hundreds = 2 hundreds.

Answer: Rs. 293

Exercise 2.5 — Add the following (pages 37-38)

(i) Yasir has 12 story books. Denar has 7 story books. How many books do they both have together?

Step 1: Add ones: 2 ones + 7 ones = 9 ones.

Step 2: Add tens: 1 ten + 0 tens = 1 ten.

Answer: 19 books together (solution given in the book; the box spells the second boy 'Dinar' while the problem text spells 'Denar')

(ii) The distance from Quetta to Muslimbagh is 108 Km and from Muslimbagh to Zhob is 194 Km. What is the distance from Quetta to Zhob?

Step 1: Add ones: 8 ones + 4 ones = 12 ones = 1 ten + 2 ones; write 2, carry 1 ten.

Step 2: Add tens: 1 ten + 0 tens + 9 tens = 10 tens = 1 hundred + 0 tens; write 0, carry 1 hundred.

Step 3: Add hundreds: 1 hundred + 1 hundred + 1 hundred = 3 hundreds.

Answer: 302 Km

(iii) A shopkeeper sold 268 books in March and 325 in April. How many books did he sell in all?

Step 1: Add ones: 8 ones + 5 ones = 13 ones = 1 ten + 3 ones; write 3, carry 1 ten.

Step 2: Add tens: 1 ten + 6 tens + 2 tens = 9 tens.

Step 3: Add hundreds: 2 hundreds + 3 hundreds = 5 hundreds.

Answer: 593 books in all

(iv) There are 365 trees of apple and 278 trees of apricot in a garden. How many trees are there in all?

Step 1: Add ones: 5 ones + 8 ones = 13 ones = 1 ten + 3 ones; write 3, carry 1 ten.

Step 2: Add tens: 1 ten + 6 tens + 7 tens = 14 tens = 1 hundred + 4 tens; write 4, carry 1 hundred.

Step 3: Add hundreds: 1 hundred + 3 hundreds + 2 hundreds = 6 hundreds.

Answer: 643 trees in all

Recall Subtraction Facts to 100 (page 38)

Subtraction of two 1-digit numbers: Salman has 9 balls and gives 3 balls to an orphan child, so 9 - 3 = 6 balls are left (subtract ones from ones).

Subtraction of a 2-digit number and a 1-digit number: 18 - 6 — Step 1, subtract ones from ones: 8 ones - 6 ones = 2 ones, write 2 in the Ones column; tens stay 1; answer 12.

Subtraction of two 2-digit numbers (page 39)

Subtract 45 from 79 — Step 1, subtract ones from ones: 9 ones - 5 ones = 4 ones, write 4 in the Ones column; Step 2, subtract tens from tens: 7 tens - 4 tens = 3 tens, write 3 in the Tens column; answer 34.

Exercise 2.6 — Subtract the following (page 39)

(1) 36 - 5

Step 1: Subtract ones: 6 ones - 5 ones = 1 one.

Step 2: Tens stay 3 tens.

Answer: 31

(2) 48 - 7

Step 1: Subtract ones: 8 ones - 7 ones = 1 one.

Step 2: Tens stay 4 tens.

Answer: 41

(3) 57 - 6

Step 1: Subtract ones: 7 ones - 6 ones = 1 one.

Step 2: Tens stay 5 tens.

Answer: 51

(4) 78 - 44

Step 1: Subtract ones: 8 ones - 4 ones = 4 ones.

Step 2: Subtract tens: 7 tens - 4 tens = 3 tens.

Answer: 34

(5) 98 - 56

Step 1: Subtract ones: 8 ones - 6 ones = 2 ones.

Step 2: Subtract tens: 9 tens - 5 tens = 4 tens.

Answer: 42

(6) 78 - 12

Step 1: Subtract ones: 8 ones - 2 ones = 6 ones.

Step 2: Subtract tens: 7 tens - 1 ten = 6 tens.

Answer: 66

Subtraction of 1-digit number from 3-digit number (page 40)

Without regrouping: subtract 4 from 127 — Step 1, subtract ones: 7 ones - 4 ones = 3 ones, write 3 in the Ones column; tens and hundreds stay 2 and 1; answer 123. With regrouping: subtract 7 from 236 — Step 1, subtract ones: 7 cannot be subtracted from 6, so borrow 1 ten as 10 ones from the tens place and carry it to the ones place; 1 ten + 6 ones = 16 ones; now 16 ones - 7 ones = 9 ones, write 9 in the Ones column; Step 2, subtract tens: 2 tens - 0 tens = 2 tens, write 2 in the Tens column; hundreds stay 2; answer 229.

Exercise 2.7 — Subtract the following (page 41)

(1) 129 - 6

Step 1: Subtract ones: 9 ones - 6 ones = 3 ones.

Step 2: Tens and hundreds stay 2 and 1.

Answer: 123

(2) 525 - 3

Step 1: Subtract ones: 5 ones - 3 ones = 2 ones.

Step 2: Tens and hundreds stay 2 and 5.

Answer: 522

(3) 579 - 8

Step 1: Subtract ones: 9 ones - 8 ones = 1 one.

Step 2: Tens and hundreds stay 7 and 5.

Answer: 571

(4) 458 - 5

Step 1: Subtract ones: 8 ones - 5 ones = 3 ones.

Step 2: Tens and hundreds stay 5 and 4.

Answer: 453

(5) 145 - 4

Step 1: Subtract ones: 5 ones - 4 ones = 1 one.

Step 2: Tens and hundreds stay 4 and 1.

Answer: 141

(6) 378 - 8

Step 1: Subtract ones: 8 ones - 8 ones = 0 ones.

Step 2: Tens and hundreds stay 7 and 3.

Answer: 370

(7) 130 - 10

Step 1: Subtract ones: 0 ones - 0 ones = 0 ones.

Step 2: Subtract tens: 3 tens - 1 ten = 2 tens.

Step 3: Hundreds stay 1.

Answer: 120

(8) 465 - 40

Step 1: Subtract ones: 5 ones - 0 ones = 5 ones.

Step 2: Subtract tens: 6 tens - 4 tens = 2 tens.

Step 3: Hundreds stay 4.

Answer: 425

(9) 843 - 30

Step 1: Subtract ones: 3 ones - 0 ones = 3 ones.

Step 2: Subtract tens: 4 tens - 3 tens = 1 ten.

Step 3: Hundreds stay 8.

Answer: 813

Subtraction of 2-digit numbers from 3-digit numbers (page 42)

Without regrouping: subtract 35 from 357 — Step 1, subtract ones: 7 ones - 5 ones = 2 ones, write 2 in the Ones column; Step 2, subtract tens: 5 tens - 3 tens = 2 tens, write 2 in the Tens column; hundreds stay 3; answer 322. With regrouping: subtract 48 from 703 — Step 1, subtract ones: 8 cannot be subtracted from 3, and zero is at the tens place, so borrow 1 hundred as 10 tens from the hundred place, then borrow 1 ten as 10 ones from the tens place and carry it to the ones place; 13 ones - 8 ones = 5 ones, write 5 in the Ones column; Step 2, subtract tens: 9 tens - 4 tens = 5 tens, write 5 in the Tens column; Step 3, subtract hundreds: 6 hundreds - 0 hundreds = 6 hundreds, write 6 in the Hundreds column; answer 655.

Exercise 2.8 — Subtract the following (page 43)

(1) 789 - 65

Step 1: Subtract ones: 9 ones - 5 ones = 4 ones.

Step 2: Subtract tens: 8 tens - 6 tens = 2 tens.

Step 3: Hundreds stay 7.

Answer: 724

(2) 184 - 42

Step 1: Subtract ones: 4 ones - 2 ones = 2 ones.

Step 2: Subtract tens: 8 tens - 4 tens = 4 tens.

Step 3: Hundreds stay 1.

Answer: 142

(3) 888 - 75

Step 1: Subtract ones: 8 ones - 5 ones = 3 ones.

Step 2: Subtract tens: 8 tens - 7 tens = 1 ten.

Step 3: Hundreds stay 8.

Answer: 813

(4) 187 - 78

Step 1: Subtract ones: 7 ones - 8 ones cannot be done, so borrow 1 ten as 10 ones; 17 ones - 8 ones = 9 ones.

Step 2: Subtract tens: 7 tens - 7 tens = 0 tens.

Step 3: Hundreds stay 1.

Answer: 109

(5) 660 - 72

Step 1: Subtract ones: 0 ones - 2 ones cannot be done, so borrow 1 ten as 10 ones; 10 ones - 2 ones = 8 ones.

Step 2: Subtract tens: 5 tens - 7 tens cannot be done, so borrow 1 hundred as 10 tens; 15 tens - 7 tens = 8 tens.

Step 3: Subtract hundreds: 5 hundreds - 0 hundreds = 5 hundreds.

Answer: 588

(6) 468 - 59

Step 1: Subtract ones: 8 ones - 9 ones cannot be done, so borrow 1 ten as 10 ones; 18 ones - 9 ones = 9 ones.

Step 2: Subtract tens: 5 tens - 5 tens = 0 tens.

Step 3: Hundreds stay 4.

Answer: 409

Subtraction of two 3-digit numbers (page 43)

Without regrouping: subtract 246 from 578 — Step 1, subtract ones: 8 ones - 6 ones = 2 ones, write 2 in the Ones column; Step 2, subtract tens: 7 tens - 4 tens = 3 tens, write 3 in the Tens column; Step 3, subtract hundreds: 5 hundreds - 2 hundreds = 3 hundreds, write 3 in the Hundreds column; answer 332. NOTE — BOOK TYPO: the heading is printed as 'Subtractiontion of two 3-digit numbers'.

Subtraction of two 3-digit numbers, with regrouping (page 44)

The worked box subtracts 426 from 785 — Step 1, subtract ones: 6 cannot be subtracted from 5, so borrow 1 ten as 10 ones from the tens place and carry it to the ones place; 1 ten + 5 ones = 15 ones; now 15 ones - 6 ones = 9 ones, write 9 in the Ones column; Step 2, subtract tens: 7 tens - 2 tens = 5 tens, write 5 in the Tens column; Step 3, subtract hundreds: 7 hundreds - 4 hundreds = 3 hundreds, write 3 in the Hundreds column; answer 359. NOTE — BOOK ERROR: the heading is printed as 'Subtract 785 from 426', which would be negative; the box itself works out 785 - 426 = 359, and that is the answer used here.

Exercise 2.9 — Subtract the following (page 44)

(1) 285 - 175

Step 1: Subtract ones: 5 ones - 5 ones = 0 ones.

Step 2: Subtract tens: 8 tens - 7 tens = 1 ten.

Step 3: Subtract hundreds: 2 hundreds - 1 hundred = 1 hundred.

Answer: 110

(2) 447 - 326

Step 1: Subtract ones: 7 ones - 6 ones = 1 one.

Step 2: Subtract tens: 4 tens - 2 tens = 2 tens.

Step 3: Subtract hundreds: 4 hundreds - 3 hundreds = 1 hundred.

Answer: 121

(3) 786 - 483

Step 1: Subtract ones: 6 ones - 3 ones = 3 ones.

Step 2: Subtract tens: 8 tens - 8 tens = 0 tens.

Step 3: Subtract hundreds: 7 hundreds - 4 hundreds = 3 hundreds.

Answer: 303

(4) 246 - 118

Step 1: Subtract ones: 6 ones - 8 ones cannot be done, so borrow 1 ten as 10 ones; 16 ones - 8 ones = 8 ones.

Step 2: Subtract tens: 3 tens - 1 ten = 2 tens.

Step 3: Subtract hundreds: 2 hundreds - 1 hundred = 1 hundred.

Answer: 128

(5) 773 - 558

Step 1: Subtract ones: 3 ones - 8 ones cannot be done, so borrow 1 ten as 10 ones; 13 ones - 8 ones = 5 ones.

Step 2: Subtract tens: 6 tens - 5 tens = 1 ten.

Step 3: Subtract hundreds: 7 hundreds - 5 hundreds = 2 hundreds.

Answer: 215

(6) 395 - 107

Step 1: Subtract ones: 5 ones - 7 ones cannot be done, so borrow 1 ten as 10 ones; 15 ones - 7 ones = 8 ones.

Step 2: Subtract tens: 8 tens - 0 tens = 8 tens.

Step 3: Subtract hundreds: 3 hundreds - 1 hundred = 2 hundreds.

Answer: 288

Real-world problem involving Subtraction (page 45)

(i) Sidra has 17 pencils, she gave 6 pencils to her brother Ali. How many pencils are left with Sidra?

Step 1: Subtract ones: 7 ones - 6 ones = 1 one.

Step 2: Tens stay 1 ten.

Answer: 11 pencils left

(iii) Ahmed had Rs. 765, he bought books for Rs. 89. How much money is left with Ahmed?

Step 1: Subtract ones: 5 ones - 9 ones cannot be done, so borrow 1 ten as 10 ones; 15 ones - 9 ones = 6 ones.

Step 2: Subtract tens: 5 tens - 8 tens cannot be done, so borrow 1 hundred as 10 tens; 15 tens - 8 tens = 7 tens.

Step 3: Subtract hundreds: 6 hundreds - 0 hundreds = 6 hundreds.

Answer: Rs. 676 left

(v) There is 445 Kg rice in a store, 278 Kg rice was given to a shopkeeper. How much rice is left in the store?

Step 1: Subtract ones: 5 ones - 8 ones cannot be done, so borrow 1 ten as 10 ones; 15 ones - 8 ones = 7 ones.

Step 2: Subtract tens: 3 tens - 7 tens cannot be done, so borrow 1 hundred as 10 tens; 13 tens - 7 tens = 6 tens.

Step 3: Subtract hundreds: 3 hundreds - 2 hundreds = 1 hundred.

Answer: 167 Kg rice left

NOTE: the book numbers these problems (i), (iii) and (v) — items (ii) and (iv) do not appear on this page.

Exercise 2.10 — Subtract the following (page 46)

(i) There are 324 birds in a zoo, if 7 are parrots then how many other birds are there?

Step 1: Subtract ones: 4 ones - 7 ones cannot be done, so borrow 1 ten as 10 ones; 14 ones - 7 ones = 7 ones.

Step 2: Subtract tens: 1 ten - 0 tens = 1 ten.

Step 3: Hundreds stay 3.

Answer: 317 other birds

(ii) A shopkeeper had 214 toys, he sold 57 toys. How many toys are left with him?

Step 1: Subtract ones: 4 ones - 7 ones cannot be done, so borrow 1 ten as 10 ones; 14 ones - 7 ones = 7 ones.

Step 2: Subtract tens: 0 tens - 5 tens cannot be done, so borrow 1 hundred as 10 tens; 10 tens - 5 tens = 5 tens.

Step 3: Subtract hundreds: 1 hundred - 0 hundreds = 1 hundred.

Answer: 157 toys left

(iii) Ariba bought 463 sweets on her birthday, she distributed 347 sweets among her friends. How many sweets are left with her?

Step 1: Subtract ones: 3 ones - 7 ones cannot be done, so borrow 1 ten as

10 ones; 13 ones - 7 ones = 6 ones.

Step 2: Subtract tens: 5 tens - 4 tens = 1 ten.

Step 3: Subtract hundreds: 4 hundreds - 3 hundreds = 1 hundred.

Answer: 116 sweets left

(iv) Muzamil had Rs. 750, he bought a cake for Rs. 425. How much money is left with Muzamil?

Step 1: Subtract ones: 0 ones - 5 ones cannot be done, so borrow 1 ten as 10 ones; 10 ones - 5 ones = 5 ones.

Step 2: Subtract tens: 4 tens - 2 tens = 2 tens.

Step 3: Subtract hundreds: 7 hundreds - 4 hundreds = 3 hundreds.

Answer: Rs. 325 left

Estimating Addition and Subtraction (page 47)

Estimation is used to find the closest value in tens or hundreds to the actual value. Estimate the sum of 32 and 17 — Step 1, round off the numbers to nearest tens: nearest ten to 32 is 30, nearest ten to 17 is 20; Step 2, add rounded numbers: 30 + 20 = 50; 50 is close to the actual value 49. Estimate the difference of 36 and 87 — Step 1, round off to nearest tens: nearest ten to 87 is 90, nearest ten to 36 is 40; Step 2, find the difference: 90 - 40 = 50; 50 is close to the actual value 51.

Exercise 2.11 — 1. Find the estimated addition to the nearest ten (page 48)

(i) 18 + 29

Step 1: Round to nearest tens: 18 rounds to 20, 29 rounds to 30.

Step 2: Add the rounded numbers: 20 + 30 = 50.

Step 3: Actual value: 18 + 29 = 47.

Answer: Estimated sum 50 (actual 47)

(ii) 42 + 37

Step 1: Round to nearest tens: 42 rounds to 40, 37 rounds to 40.

Step 2: Add the rounded numbers: 40 + 40 = 80.

Step 3: Actual value: 42 + 37 = 79.

Answer: Estimated sum 80 (actual 79)

Exercise 2.11 — 2. Find the estimated difference to the nearest ten (page 48)

(i) 79 - 35

Step 1: Round to nearest tens: 79 rounds to 80, 35 rounds to 40.

Step 2: Subtract the rounded numbers: 80 - 40 = 40.

Step 3: Actual value: 79 - 35 = 44.

Answer: Estimated difference 40 (actual 44)

(ii) 92 - 34

Step 1: Round to nearest tens: 92 rounds to 90, 34 rounds to 30.

Step 2: Subtract the rounded numbers: 90 - 30 = 60.

Step 3: Actual value: 92 - 34 = 58.

Answer: Estimated difference 60 (actual 58)

Exercise 2.11 — 3. Tick the correct answer (page 48)

(i) 16 + 18 is estimated to:

20

30

✓ 40

(ii) 37 + 12 is estimated to:

✓ 50

40

30

(iii) 43 + 31 is estimated to:

50

60

✓ 70

(iv) 49 - 38 is estimated to: NOTE — BOOK ERROR: the printed options (80, 90, 100) do not include the correct estimated difference. Rounding to nearest tens, 49 rounds to 50 and 38 rounds to 40, so the estimated difference is 50 - 40 = 10 (actual 11). No option is ticked.

(v) 21 - 18 is estimated to: NOTE — BOOK ERROR: the printed options (20, 30, 40) do not include the correct estimated difference. Rounding to nearest tens, 21 rounds to 20 and 18 rounds to 20, so the estimated difference is 20 - 20 = 0 (actual 3). No option is ticked.

I have Learnt (page 49)

Addition of numbers mentally and in written form without regrouping; addition of 3-digit numbers and 1-digit number; addition of 3-digit numbers and tens; addition of 3-digit numbers and 2-digit numbers; addition of 3-digit numbers; solving real-world problems involving addition; subtraction of 1-digit number from 3-digit number; subtraction of tens from 3-digit numbers; subtraction of 2-digit numbers from 3-digit numbers; subtraction of 3-digit numbers from 3-digit numbers; estimation of addition and subtraction. Vocabulary used: Addition, sum, equals,

subtraction, left, difference, money, ones, tens, hundreds, re-grouping, estimate, nearest.

Review Exercise — 1. Tick the correct answer (page 49)

(i) 15 + 1 =

- 7
- 14
- ✓ 16
- 25

(ii) 10 + 0 =

- 1
- ✓ 10
- 100
- 101

(iii) 354 - 3 =

- 54
- 324
- 331
- ✓ 351

(iv) When we add two 3-digit numbers, add first:

- ✓ Ones
- Tens
- Hundreds
- Thousands

(v) When zero is added to a number, the answer is:

- Zero
- ✓ Number itself

Next to number

Small number

Review Exercise — 2. Add / subtract the following (page 50)

(i) 385 + 4

Step 1: Add ones: 5 ones + 4 ones = 9 ones.

Step 2: Tens and hundreds stay 8 and 3.

Answer: 389

(ii) 825 - 13

Step 1: Subtract ones: 5 ones - 3 ones = 2 ones.

Step 2: Subtract tens: 2 tens - 1 ten = 1 ten.

Step 3: Hundreds stay 8.

Answer: 812

Review Exercise — 3. Solve the following (page 50)

(i) 165 - 43

Step 1: Subtract ones: 5 ones - 3 ones = 2 ones.

Step 2: Subtract tens: 6 tens - 4 tens = 2 tens.

Step 3: Hundreds stay 1.

Answer: 122

(ii) 534 + 346

Step 1: Add ones: 4 ones + 6 ones = 10 ones = 1 ten + 0 ones; write 0, carry 1 ten.

Step 2: Add tens: 1 ten + 3 tens + 4 tens = 8 tens.

Step 3: Add hundreds: 5 hundreds + 3 hundreds = 8 hundreds.

Answer: 880

Review Exercise — 4. Find the estimated addition (page 50)

18 + 13

Step 1: Round to nearest tens: 18 rounds to 20, 13 rounds to 10.

Step 2: Add the rounded numbers: 20 + 10 = 30.

Step 3: Actual value: 18 + 13 = 31.

Answer: Estimated sum 30 (actual 31)

37 + 41

Step 1: Round to nearest tens: 37 rounds to 40, 41 rounds to 40.

Step 2: Add the rounded numbers: 40 + 40 = 80.

Step 3: Actual value: 37 + 41 = 78.

Answer: Estimated sum 80 (actual 78)

28 - 14

Step 1: Round to nearest tens: 28 rounds to 30, 14 rounds to 10.

Step 2: Subtract the rounded numbers: 30 - 10 = 20.

Step 3: Actual value: 28 - 14 = 14.

Answer: Estimated difference 20 (actual 14)

Review Exercise — 5. Hammad buys a toy for Rs. 175 and a doll for Rs. 35, what amount did Hammad spend altogether? (page 50)

175 + 35

Step 1: Add ones: 5 ones + 5 ones = 10 ones = 1 ten + 0 ones; write 0, carry 1 ten.

Step 2: Add tens: 1 ten + 7 tens + 3 tens = 11 tens = 1 hundred + 1 ten; write 1, carry 1 hundred.

Step 3: Add hundreds: 1 hundred + 1 hundred + 0 hundreds = 2 hundreds.

Answer: Rs. 210 altogether

Review Exercise — 6. There are 138 cows and 235 sheep in a farm, how many animals are there in the farm? (page 51)

138 + 235

Step 1: Add ones: 8 ones + 5 ones = 13 ones = 1 ten + 3 ones; write 3, carry 1 ten.

Step 2: Add tens: 1 ten + 3 tens + 3 tens = 7 tens.

Step 3: Add hundreds: 1 hundred + 2 hundreds = 3 hundreds.

Answer: 373 animals in the farm

Review Exercise — 7. Amna has 38 toys, she distributed 25 among her friends, how many toys are left with her? (page 51)

38 - 25

Step 1: Subtract ones: 8 ones - 5 ones = 3 ones.

Step 2: Subtract tens: 3 tens - 2 tens = 1 ten.

Answer: 13 toys left

Review Exercise — 8. Yasir got Rs. 345 as Eidi. He gave Rs. 160 to his sister. How much amount is left with Yasir? (page 51)

345 - 160

Step 1: Subtract ones: 5 ones - 0 ones = 5 ones.

Step 2: Subtract tens: 4 tens - 6 tens cannot be done, so borrow 1 hundred as 10 tens; 14 tens - 6 tens = 8 tens.

Step 3: Subtract hundreds: 2 hundreds - 1 hundred = 1 hundred.

Answer: Rs. 185 left

Unit 3 — Multiplication and Division (pages 52 to 79)

Students' learning outcomes (page 52): After completing this unit, students will be able to: recognize even and odd numbers; count and write in 3s, 4s, 5s, 10s and 100s; recognize counting in 3s, 4s as multiplication table of three and four; recognize multiplication as repeated addition and develop multiplication (times table) table for 2, 3, 4, 5 and 10; write multiplication statements (i.e. sentences) using concrete and pictorial representation; recognize using concrete and pictorial representation that the multiplication of any two numbers can be done in any order; multiply mentally and in written form using the multiplication tables that they know: 1-digit by another 1-digit number; multiply 2-digit number by a 1-digit number using multiplication grid; multiply a number with 0 and 1; solve real world problems involving multiplications; recognize division as repeated subtraction through concrete and pictorial representation; write division statements (i.e. sentences) using concrete and pictorial representation; recognize that division cannot be done in any order, which can be shown using concrete objects and pictorial representations; divide mentally and in written form: 1-digit number by another 1-digit number without reminder, 2-digit

number by a 1-digit number (without reminder). Opener facts: 3 x 6 = 18; 18 / 3 = 6; 6 x 3 = 18; 18 / 6 = 3.

What are the even and odd numbers? (page 53)

Count the balls in pairs of 2 to understand even and odd numbers. (1) One pair + one single ball = 3 balls — odd. (2) Three pairs = 6 balls — even. (3) Three pairs + one single ball = 7 balls — odd. (4) Four pairs = 8 balls — even. (5) Four pairs + one single ball = 9 balls — odd. The numbers of balls which are in pairs are called even numbers, and the balls that are out of pairs are called odd numbers. Key point: the numbers divisible by 2 are called even numbers; the numbers not divisible by 2 are called odd numbers.

Even and Odd Numbers (page 54)

Look at the pictures. There are 6 apples in the plate. We can make groups of 2 apples. 6 is an even number. There are 5 apples in the plate. When we make groups of 2 apples, one apple is left. A group cannot be made. 5 is an odd number. Even numbers: numbers that can be grouped into two. Example: 0, 2, 4, 6, 8, 10, 12, 14, etc. Odd numbers: numbers that cannot be grouped into two. Example: 1, 3, 5, 7, 9, 11, 13, 15, etc.

Counting in Steps (pages 55-56)

Counting in 2s: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20 (adding 2 each time). Counting in 3s: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30 (adding 3 each time). Counting in steps of 4: 4, 8, 12, 16, 20, 24, 28, 32, 36, 40 (adding 4 each time). Counting in 5s: 5, 10, 15, 20, 25, 30, 35, 40, 45, 50 (adding 5 each time). Counting in 10s: 10, 20, 30, 40, 50, 60, 70, 80, 90, 100 (adding 10 each time). Counting in 100s: 100, 200, 300, 400, 500, 600, 700, 800,

900 (adding 100 each time).

Exercise 3.1 — 1. Circle the odd numbers (page 57)

Q: From 10, 11, 44, 33, 88, 15, 99, the odd numbers are:

A: 11, 33, 15, 99 (10, 44 and 88 are even).

Exercise 3.1 — 2. Circle the even numbers (page 57)

Q: From 1, 20, 33, 44, 50, 61, 73, 81, 96, the even numbers are:

A: 20, 44, 50, 96 (1, 33, 61, 73 and 81 are odd).

Exercise 3.1 — 3. Complete counting of the given steps (page 57)

Chain — complete sequence

(i) Counting in 2s — 2, 4, 6, 8, 10, 12, 14, 16, 18, 20

(ii) Counting in 3s — 3, 6, 9, 12, 15, 18, 21, 24, 27, 30

(iii) Counting in 4s — 4, 8, 12, 16, 20, 24, 28, 32, 36, 40

(iv) Counting in 5s — 35, 40, 45, 50, 55, 60, 65, 70, 75, 80

(v) Counting in 10s — 10, 20, 30, 40, 50, 60, 70, 80, 90, 100

Multiplication as Repeated Addition (page 58)

Let us add: 2 + 2 + 2 + 2 = 8 (four groups of 2 stars). 4 times 2 = 8, or 2 x 4 = 8. The process of repeated addition is called Multiplication. Key point: the symbol 'x' is for multiplication. Next: 3 + 3 + 3 + 3 + 3 = 15 (five bunches of 3 flowers). 5 times 3 = 15, or 3 x 5 = 15. Next: 4 + 4 + 4 + 4 + 4 = 20 (five plates of 4 apples). 5 times 4 = 20, or 4 x 5 = 20.

Multiplication Tables — 2 Times Table (page 59)

Table — product

1 x 2 — 2

2 x 2 — 4

3 x 2 — 6

4 x 2 — 8

5 x 2 — 10

6 x 2 — 12

7 x 2 — 14

8 x 2 — 16

9 x 2 — 18

10 x 2 — 20

NOTE — BOOK TYPO: every row of the 2 Times Table is labelled '1 times 2' underneath, even for 2 x 2, 3 x 2 and so on; the products themselves are correct.

Multiplication Tables — 3 Times Table (page 60)

Table — product

1 x 3 — 3

2 x 3 — 6

3 x 3 — 9

4 x 3 — 12

5 x 3 — 15

6 x 3 — 18

7 x 3 — 21

8 x 3 — 24

9 x 3 — 27

10 x 3 — 30

Multiplication Tables — 4 Times Table (page 61)

Table — product

1 x 4 — 4

2 x 4 — 8

3 x 4 — 12

4 x 4 — 16

5 x 4 — 20

6 x 4 — 24

7 x 4 — 28

8 x 4 — 32

9 x 4 — 36

10 x 4 — 40

Multiplication Tables — 5 Times Table (page 62)

1 dice = 5 dots.

Table — product

1 x 5 — 5

2 x 5 — 10

3 x 5 — 15

4 x 5 — 20

5 x 5 — 25

6 x 5 — 30

7 x 5 — 35

8 x 5 — 40

9 x 5 — 45

10 x 5 — 50

Multiplication Tables — 10 Times Table (page 63)

2 dices = 10 dots.

Table — product

1 x 10 — 10

2 x 10 — 20

3 x 10 — 30

4 x 10 — 40

5 x 10 — 50

6 x 10 — 60

7 x 10 — 70

8 x 10 — 80

9 x 10 — 90

10 x 10 — 100

Exercise 3.2 — 1. Fill in the blanks (page 64)

(i) Four groups of 3 candles: 3 + 3 + 3 + 3

Step 1: Add: 3 + 3 = 6, 6 + 3 = 9, 9 + 3 = 12.

Answer: 12; 3 x 4 = 12

(ii) Five dice of 5 dots: 5 + 5 + 5 + 5 + 5

Step 1: Add: 5 + 5 = 10, 10 + 5 = 15, 15 + 5 = 20, 20 + 5 = 25.

Answer: 25; 5 x 5 = 25

(iii) Five pairs of mangoes: 2 + 2 + 2 + 2 + 2

Step 1: Add: 2 + 2 = 4, 4 + 2 = 6, 6 + 2 = 8, 8 + 2 = 10.

Answer: 10; 2 x 5 = 10

Exercise 3.2 — 2. Complete the following (page 64)

Product — repeated addition

(i) 2 x 2 (given) — 2 + 2 = 4

(ii) 3 x 2 — 3 + 3 = 6

(iii) 5 x 4 — 5 + 5 + 5 + 5 = 20

(iv) 4 x 5 — 4 + 4 + 4 + 4 + 4 = 20

(v) 10 x 3 — 10 + 10 + 10 = 30

(vi) 2 x 4 — 2 + 2 + 2 + 2 = 8

The book's boxes use the first factor as the addend and the second factor as the number of addends, so 10 x 3 is written as three 10s and 2 x 4 as four 2s.

Exercise 3.2 — 3. Match the following (page 65)

Product — answer

(i) 2 x 3 — 6

(ii) 3 x 3 — 9

(iii) 4 x 5 — 20

(iv) 3 x 10 — 30

(v) 10 x 4 — 40

(vi) 5 x 7 — 35

Exercise 3.2 — 4. Complete the multiplication table wheels (page 65)

(i) Centre 3 wheel: outer ring = 3 times the inner number — 1 gives 3 (given), 2 gives 6, 3 gives 9, 4 gives 12 (given), 5 gives 15, 6 gives 18, 7 gives 21 (given), 8 gives 24, 9 gives 27 (given), 10 gives 30. (ii) Centre 5 wheel: outer ring = 5 times the inner number — 1 gives 5 (given), 2 gives

10, 3 gives 15, 4 gives 20, 5 gives 25 (given), 6 gives 30, 7 gives 35, 8 gives 40, 9 gives 45, 10 gives 50.

Multiplication Grid (page 66)

To find the product of two numbers, locate the row and column where they intersect on the multiplication table. For example, the product of 5 and 4 is 20, as shown in the table (row 4, column 5 is highlighted as 20). Multiplication of a 2-digit number by a 1-digit number using the multiplication grid: multiply 10 and 4 — use the grid, circle 10 in the first column and 4 in the first row; the number 40 below 10 in the fifth row is the answer, so 10 x 4 = 40.

Multiplication of a number by 0 and 1 (page 67)

When we multiply a number with zero, then the product is always zero. 0 x 4 means 0 times 4 = 0, and 4 x 0 means 4 times 0 = 0. Examples: 5 x 0 = 0; 7 x 0 = 0; 3 x 0 = 0; 0 x 2 = 0; 0 x 8 = 0; 0 x 5 = 0. If we multiply a number with 1, we always get the number itself. 1 x 5 means 1 time 5 = 5, and 5 x 1 means 5 times 1 = 5. Examples: 3 x 1 = 3; 5 x 1 = 5; 2 x 1 = 2; 4 x 1 = 4; 1 x 2 = 2; 1 x 3 = 3; 13 x 1 = 13; 10 x 1 = 10; 1 x 10 = 10.

Real-world problem involving multiplication (page 67)

A bicycle has 2 wheels. How many wheels do 5 bicycles have?

Step 1: Wheels of 1 bicycle = 2.

Step 2: Wheels of 5 bicycles = 5 times 2 = 5 x 2.

Answer: 10 wheels

Order of Multiplication (page 68)

Let us multiply 3 x 4. We have 3 groups of 4 bananas: 4 + 4 + 4 = 12, so 3 x 4 = 12. Now multiply 4 x 3: 4 groups of 3 bananas: 3 + 3 + 3 + 3 = 12, so 4 x 3 = 12. Hence 3 x 4 = 4 x 3. Multiplication can be done in any order.

Exercise 3.3 — 1. Multiply the following (page 68)

Product — answer

(1) 5 x 2 — 10

(2) 10 x 8 — 80

(3) 5 x 4 — 20

(4) 6 x 5 — 30

(5) 4 x 3 — 12

(6) 10 x 5 — 50

(7) 8 x 4 — 32

(8) 7 x 3 — 21

Exercise 3.3 — 2. Fill in the blanks (page 69)

Product — answer

(i) 8 x 0 — 0

(ii) 10 x 0 — 0

(iii) 5 x 1 — 5

(iv) 12 x 1 — 12

(v) 7 x 0 — 0

(vi) 0 x 5 — 0

(vii) 9 x 1 — 9

(viii) 1 x 13 — 13

Exercise 3.3 — 3. Fill in the blanks (order of multiplication, page 69)

Statement — completed

(i) 3 x 2 = 2 x ___ (given) — 3

(ii) 10 x 1 = 1 x ___ — 10

(iii) 5 x 4 = ___ x 5 — 4

(iv) 3 x 5 = 5 x ___ — 3

Exercise 3.3 — 4. Dinar has 4 friends, he gave Rs. 5 to each. How much money did he distribute? (page 69)

4 friends x Rs. 5 each

Step 1: Number of friends = 4.

Step 2: Money given to each = Rs. 5.

Step 3: Total money = 4 x 5 = 20.

Answer: Rs. 20 (total given in the book)

Exercise 3.3 — 5. There are 5 books in one bag. How many books are there in 6 bags? (page 69)

5 books in one bag x 6 bags

Step 1: Books in one bag = 5.

Step 2: Number of bags = 6.

Step 3: Total books = 5 x 6 = 30.

Answer: 30 books

Exercise 3.3 — 6. The price of one pencil is Rs. 10. What is the

price of 5 pencils? (page 69)

Rs. 10 x 5 pencils

Step 1: Price of 1 pencil = Rs. 10.

Step 2: Number of pencils = 5.

Step 3: Total price = 10 x 5 = 50.

Answer: Rs. 50

Exercise 3.3 — 7. There are 5 flowers in a flowerpot. How many flowers are there in 3 flowerpots? (page 70)

5 flowers in each pot x 3 flowerpots

Step 1: Number of flowerpots = 3.

Step 2: Number of flowers in each = 5.

Step 3: Total flowers = 3 x 5 = 15.

Answer: 15 flowers (total given in the book)

Division as Repeated Subtraction (page 70)

Jamil has 8 balls; he wanted to distribute them equally among his 4 friends. He gives two balls to each friend: 8 - 2 - 2 - 2 - 2 = 0, shown as 8, 6, 4, 2, 0 with -2 steps. Each friend gets 2 balls. Step by step: 8 - 2 = 6; 6 - 2 = 4; 4 - 2 = 2; 2 - 2 = 0. Key point: the process of repeated subtraction is called division. The symbol for division is the division sign. We can write it as 8 divided by 4 = 2, and read it as '8 divided by 4 is equal to 2'. In long form: 4 into 8 goes 2 times, 2 x 4 = 8, remainder 0.

Division of 1-digit and 2-digit number by 1-digit number (page 71)

Aliya has 9 pencils; she wants to put them equally in 3 boxes. She puts 1 pencil in each box. It means 3 can be subtracted 3 times from 9: 9 - 3 = 6; 6 - 3 = 3; 3 - 3 = 0. So 9 - 3 - 3 - 3 = 0, and 9 divided by 3 = 3, or in long form 3 into 9 goes 3 times, 3 x 3 = 9, remainder 0. Key point: division means equal sharing or equal grouping. Dividing 6 balloons in 3 groups: 6 - 3 = 3; 3 - 3 = 0, so 6 divided by 3 = 2, or in long form 3 into 6 goes 2 times, 2 x 3 = 6, remainder 0.

Division examples (page 72)

Example 1: Divide 9 by 3 — 9 divided by 3 = 3; in long form 3 into 9 goes 3 times, 3 x 3 = 9, remainder 0. Example 2: Divide 15 by 3 — 15 divided by 3 = 5; in long form 3 into 15 goes 5 times, 5 x 3 = 15, remainder 0. Example 3: Divide 24 by 4 — 24 divided by 4 = 6; in long form 4 into 24 goes 6 times, 6 x 4 = 24, remainder 0. Example 4: Divide 35 by 5 — 35 divided by 5 = 7; in long form 5 into 35 goes 7 times, 7 x 5 = 35, remainder 0.

Order in Division (pages 72-73)

Let us divide 4 cakes among 2 children equally: 4 divided by 2 = 2; each child will get 2 cakes. Now we change the order and divide 2 cakes among 4 children: each child will get half a cake, so 2 divided by 4 = half a cake (1/2). Hence 4 divided by 2 is not equal to 2 divided by 4. Therefore, division of one number by another number cannot be done in any order.

Exercise 3.4 — 1. Divide 10 mangoes equally in 2 plates by using repeated subtraction (page 73)

10 mangoes in 2 plates

Step 1: 10 - 2 = 8.

Step 2: 8 - 2 = 6.

Step 3: 6 - 2 = 4.

Step 4: 4 - 2 = 2.

Step 5: 2 - 2 = 0.

Step 6: 2 was subtracted 5 times.

Answer: 10 divided by 2 = 5 (5 mangoes in each plate)

Exercise 3.4 — 2. Divide 18 pencils equally in 3 children (page 74)

18 pencils among 3 children

Step 1: 18 - 3 = 15 (given).

Step 2: 15 - 3 = 12.

Step 3: 12 - 3 = 9.

Step 4: 9 - 3 = 6.

Step 5: 6 - 3 = 3.

Step 6: 3 - 3 = 0.

Step 7: 3 was subtracted 6 times.

Answer: 18 divided by 3 = 6 (6 pencils for each child)

Exercise 3.4 — 3. Look at the picture and fill in the boxes (page 74)

Q: Total number of balls (three groups of 5 balls):

A: 15.

Q: Number of groups:

A: 3.

Q: Balls in each group:

A: 5.

Q: Division statement:

A: 15 divided by 3 = 5.

Exercise 3.4 — 4. Complete the division using multiplication table (page 74)

Division — answer

(i) 8 divided by 4 — 2

(ii) 10 divided by 2 — 5

(iii) 12 divided by 3 — 4

(iv) 20 divided by 4 — 5

(v) 25 divided by 5 — 5

(vi) 28 divided by 4 — 7

(vii) 50 divided by 10 — 5

(viii) 35 divided by 5 — 7

Exercise 3.4 — 5. Solve the following (page 75)

Long division — answer

(i) 15 divided by 3 (given) — 5, remainder 0

(ii) 18 divided by 3 — 6, remainder 0

(iii) 16 divided by 2 — 8, remainder 0

(iv) 40 divided by 5 — 8, remainder 0

(v) 32 divided by 4 — 8, remainder 0

(vi) 15 divided by 5 — 3, remainder 0

Exercise 3.4 — 6. There are 45 students in class, the teacher divides them into groups of 5 students. How many groups are made? (page 75)

45 students in groups of 5

Step 1: Number of students = 45.

Step 2: Each group has students = 5.

Step 3: Number of groups made = 45 divided by 5.

Answer: 9 groups

Exercise 3.4 — 7. There are 4 bottles in a carton. How many cartons are needed for 36 bottles? (page 75)

36 bottles, 4 bottles in a carton

Step 1: Number of bottles = 36.

Step 2: Bottles in one carton = 4.

Step 3: Number of cartons needed = 36 divided by 4.

Answer: 9 cartons

NOTE: the book's side boxes for item 7 reuse the labels 'Number of students' and 'Each group has students' from item 6, although the problem is about bottles and cartons; the bottle wording is used here.

I have Learnt (page 75)

Difference between even and odd number; counting in steps of 3s, 4s, 5s, 10s and 100s; multiplication is the process of repeated addition and its symbol is 'x'; multiplication tables of 2, 3, 4, 5, and 10; writing multiplication sentences; multiplication of 1-digit number by 1-digit number; multiplication of 2-digit number by 1-digit number; when a

number is multiplied by zero (0), the answer is always zero; when a number is multiplied by one (1), the answer is always the number itself; multiplication of 2 numbers can be done in any order; division means equal distribution — it is the process of repeated subtraction, its symbol is the division sign; division of one number by another number cannot be done in any order.

Review Exercise — 1. Tick the correct answer (page 76)

(i) The multiplication sentence for 3 + 3 + 3 + 3 is:

- ✓ 3 x 4
- 4 x 3
- 4 x 4
- 3 x 3

(ii) The repeated addition for 2 x 5 is:

- 2 x 2
- 5 x 5
- ✓ 5 + 5
- 2 + 2

(iii) Complete the number sentence 4 + 4 + 4 + 4 + 4 + 4:

- ✓ 6 times 4
- 4
- 18
- 20

(iv) 5 times 6 is equal to:

- 11
- 25
- ✓ 30

36

(v) How many fingers are there in five hands:

15

20

✓ 25

30

(vi) 5 divided by 5 =

0

✓ 1

5

25

(vii) 50 divided by 5 =

5

✓ 10

50

55

(viii) 8 - 2 - 2 - 2 - 2 as a division sentence is:

✓ 8 divided by 2

8 divided by 4

4 divided by 2

8 divided by 8

NOTES on Review 1: (i) 3 x 4 and 4 x 3 are both 12; the book's own pattern (four groups of 3 written as 3 x 4 on page 58) gives 3 x 4, which is ticked. (iii) Six 4s make 6 times 4, whose product is 24; 24 is not among the printed options, so the sentence form '6 times 4' is ticked. (viii) Subtracting 2 four times from 8 reaches 0, so the division sentence is 8 divided by 2; its quotient is 4.

Review Exercise — 2. Divide 6 balls in 2 teams equally (page 77)

Q: 6 balls in 2 teams:

A: 6 divided by 2 = 3 (3 balls in each team).

Review Exercise — 3. Colour the even numbers in red and the odd numbers in blue (page 77)

Q: From 8, 11, 24, 27, 32, 49, 50, 56, 60, 61, 70 — even (red):

A: 8, 24, 32, 50, 56, 60, 70.

Q: From the same list — odd (blue):

A: 11, 27, 49, 61.

Review Exercise — 4. Complete the counting in 5s (page 77)

Q: Counting in 5s (given 5, 25, 40):

A: 5, 10, 15, 20, 25, 30, 35, 40.

Review Exercise — 5. Express using multiplication statements (page 77)

Repeated addition — multiplication statement

(i) 4 + 4 + 4 + 4 + 4 (given) — 5 times 4; 5 x 4 = 20

(ii) 3 + 3 + 3 + 3 + 3 + 3 — 6 times 3; 3 x 6 = 18

(iii) 10 + 10 + 10 + 10 + 10 + 10 — 6 times 10; 10 x 6 = 60

(iv) 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 — 8 times 2; 2 x 8 = 16

Review Exercise — 6. Complete the following tables (wheels, page 77)

(i) Centre 2 wheel: outer ring = 2 times the inner number — 1 gives 2

(given), 2 gives 4 (given), 3 gives 6, 4 gives 8, 5 gives 10, 6 gives 12, 7 gives 14, 8 gives 16, 9 gives 18, 10 gives 20. (ii) Centre 4 wheel: outer ring = 4 times the inner number — 1 gives 4 (given), 2 gives 8 (given), 3 gives 12, 4 gives 16, 5 gives 20, 6 gives 24, 7 gives 28, 8 gives 32, 9 gives 36, 10 gives 40.

Review Exercise — 7. Fill in the boxes (page 78)

Statement — answer

(i) 9 x 2 (given) — 18

(ii) 7 x 3 — 21

(iii) 12 divided by 4 — 3

(iv) 27 divided by 3 — 9

(v) 4 x ___ = 24 — 6

(vi) ___ x 5 = 35 — 7

(vii) 40 divided by 5 — 8

(viii) 30 divided by 10 — 3

Review Exercise — 8. Solve the following (page 78)

Product — answer

(i) 5 x 3 (given) — 15

(ii) 6 x 0 — 0

(iii) 9 x 2 — 18

(iv) 7 x 2 — 14

Review Exercise — 9. Write in division sentences (page 78)

Repeated subtraction — division sentence

(i) 8 - 2 - 2 (given) — 8 divided by 2 = 4

(ii) 8 - 2 - 2 - 2 - 2 — 8 divided by 2 = 4

(iii) 9 - 3 - 3 - 3 — 9 divided by 3 = 3

(iv) 12 - 3 - 3 - 3 - 3 — 12 divided by 3 = 4

NOTE: item (i) prints only two subtractions (8 - 2 - 2 = 4, not 0) but gives the division sentence 8 divided by 2 = 4; the full repeated-subtraction chain for 8 divided by 2 is item (ii).

Review Exercise — 10. Solve the following (page 78)

Long division — answer

(i) 18 divided by 2 (given) — 9, remainder 0

(ii) 32 divided by 4 — 8, remainder 0

(iii) 27 divided by 3 — 9, remainder 0

Review Exercise — 11. There are 7 days in a week. How many days are there in 4 weeks? (page 79)

7 days in 1 week x 4 weeks

Step 1: Days in 1 week = 7.

Step 2: Days in 4 weeks = 7 x 4.

Answer: 28 days

Review Exercise — 12. 40 flowers have to be put in 5 vases. How many flowers should be put in each vase? (page 79)

40 flowers in 5 vases

Step 1: Total number of flowers = 40.

Step 2: Number of vases = 5.

Step 3: Flowers in each vase = 40 divided by 5.

Answer: 8 flowers in each vase

Review Exercise — 13. A cake has 12 slices, Hira wants to divide it between 4 persons. How many slices will each person get? (page 79)

12 slices among 4 persons

Step 1: Total number of slices = 12.

Step 2: Number of people = 4.

Step 3: Each person gets = 12 divided by 4.

Answer: 3 slices each

Unit 4 — Money (pages 80 to 88)

Students' learning outcomes (page 80): After completing this unit, students will be able to: identify international currency and denominations (for instance dollar); solve money-related problems using addition and subtraction with Pakistani currency and a few selected international notes, such as the dollar. The opener shows Pakistani notes (Rs. 10, 20, 50, 100, 500), US dollar notes and coins.

Pakistani Currency (page 81)

We purchase different items to meet our daily needs and use money to pay for them. Pakistani currency comes in both coins and notes. Notes we commonly use (all labelled in the book): Rs. 10 note, Rs. 20 note, Rs. 50

note, Rs. 100 note, Rs. 500 note, Rs. 1000 note. Coins we commonly use: the pictured coins include the Rs. 5 coin and the Rs. 10 coin (values legible on the coins) and smaller silver coins.

International Currencies (page 82)

American currency: 1 Dollar = 100 Cents. Pictured: 1 Cent coin, 1 Dollar note, 10 Dollars note. British currency: 1 Pound = 100 Pennies. Pictured: 1 Penny coin, 1 Pound note, 10 Pounds note. Saudi currency: pictured 500 Riyals note and coins of 2 Riyals, 1 Riyal, 50 Halalah, 25 Halalah, 5 Halalah and 1 Riyal.

Ways of exchanging money (page 83)

Different ways of exchanging money: 10 Pounds = Ten Penny Coins (as printed — see note below); 20 Pounds = Two notes of Ten Pounds; 1 Dollar = Hundred cents; 10 Dollars = Ten notes of one dollar. Example: count the notes/coins and find the total amount — a 10-dollar note + a 1-dollar note + a 1-cent coin are added as $10 + $1 + $1 = $12 in the book. NOTE — BOOK QUIRK: the example counts the 1-cent coin as $1; at real value the total would be $11.01, but the book's printed total is $12 and that is what is used for the exercise below. Also, the '10 Pounds = Ten Penny Coins' row is misprinted as written (ten penny coins make 10 pennies, not 10 pounds); the intended meaning is ten 1-pound coins, but it is reproduced here as printed and flagged.

Exercise 4.1 — 1. The price of a clock is $12. Tick on dollar notes and coins for its price (page 83)

Q: To make $12 from the pictured $10 note, $1 note and two 1-cent coins:

A: Tick the $10 note, the $1 note and ONE cent coin. Following the book's own example on this page, the cent coin is counted as $1, giving $10 + $1 + $1 = $12. At real value this would be $11.01, so the book's counting convention is flagged here.

Exercise 4.1 — 2. The price of a mug is 20 pounds. Tick on pound notes and coins for its price (page 84)

Q: To make 20 pounds from the pictured two 10-pound notes and two 1-penny coins:

A: Tick BOTH 10-pound notes (10 + 10 = 20). Do not tick the penny coins.

Addition and Subtraction of money (page 84)

Yasir has Rs. 50 and Aslam has Rs. 20. What is the total amount they both have? Add the amount written on both currency notes: Rs. 50 + Rs. 20 = Rs. 70. They have Rs. 70. Example: count the notes/coins and find the total amount — a Rs. 20 note + a Rs. 10 note + a Rs. 5 coin: Rs. 20 + Rs. 10 + Rs. 5 = Rs. 35.

Money change example (page 85)

Amjad bought a chocolate for Rs. 30, he gave the shopkeeper a 50-rupee note. How much amount will the shopkeeper return?

Step 1: Rupees given to shopkeeper = Rs. 50.

Step 2: Price of chocolate = Rs. 30.

Step 3: Return amount = Rs. 50 - Rs. 30.

Answer: Rs. 20

Exercise 4.2 — 1. Count the money and write the total value

(page 85)

(i) Three Rs. 5 coins and one silver coin

Step 1: 5 + 5 + 5 = 15.

Step 2: Add the silver coin, read as Rs. 2 (its denomination is small in print; the same building-design coin appears in the page 81 coin row).

Step 3: 15 + 2 = 17.

Answer: Rs. 17

(ii) Rs. 20 note, Rs. 10 note, Rs. 5 coin and one silver coin

Step 1: 20 + 10 = 30.

Step 2: 30 + 5 = 35.

Step 3: 35 + 2 = 37 (silver coin read as Rs. 2, as in item (i)).

Answer: Rs. 37

(iii) Rs. 50 note, Rs. 20 note, Rs. 10 note and two Rs. 5 coins

Step 1: 50 + 20 = 70.

Step 2: 70 + 10 = 80.

Step 3: 80 + 5 + 5 = 90.

Answer: Rs. 90

Exercise 4.2 — 2. Add and find the total cost of the given object (pages 85-86)

(i) Toy car (10 pounds) and ball (20 pounds)

Step 1: 10 + 20 = 30.

Answer: 30 pounds

(ii) School bag (Rs. 65) and doll (Rs. 85)

Step 1: 65 + 85 = 150.

Answer: Rs. 150

(iii) Pencil (Rs. 40) and book (Rs. 55)

Step 1: 40 + 55 = 95.

Answer: Rs. 95

Exercise 4.2 — 3. Haris had Rs. 50, he bought a book for Rs. 42. How much money was returned? (page 86)

Rs. 50 - Rs. 42

Step 1: Haris had amount = Rs. 50.

Step 2: Price of book = Rs. 42.

Step 3: Return amount = 50 - 42 = 8.

Answer: Rs. 8

Exercise 4.2 — 4. Ajmal had 175 Riyal, he bought a pizza for 145 Riyal. How much money is left with Ajmal? (page 86)

175 Riyal - 145 Riyal

Step 1: Ajmal had money = 175 Riyal.

Step 2: Price of pizza = 145 Riyal.

Step 3: Amount left = 175 - 145 = 30.

Answer: 30 Riyal

Exercise 4.2 — 5. Nimra bought a bag for 13 dollars and a purse for 6 dollars. How much money did she spend? (page 87)

13 dollars + 6 dollars

Step 1: Price of bag = $13.

Step 2: Price of purse = $6.

Step 3: Money spent = 13 + 6 = 19.

Answer: $19

I have Learnt (page 87)

Identification of international currency notes and coins; same denomination of different currency notes and coins; identification of international currency notes (Dollar, pound) and Riyal; denomination of Dollar (10 dollar = 10 notes of 1 dollar); addition and subtraction of money problems involving Pakistani and international currency notes. Vocabulary used: Currency, notes, coins, money, international, dollar, pound, riyal, denomination.

Review Exercise — 1. Choose the correct option (pages 87-88)

(i) How many 1 dollar coins make 20 dollars?

- 4
- 5
- 6
- ✓ 20

(ii) How many 10 dollar notes make 100 dollars?

- 5
- 8
- ✓ 10
- 20

(iii) 1 dollar = ___ Cents:

- 10
- 20

50

✓ 100

(iv) Rs. 15 + Rs. 25 = Rs. ___:

✓ 40

25

50

40

NOTE on Review 1 (iv): options (a) and (d) both print 40; either tick gives the correct sum, Rs. 40.

Review Exercise — 2. Add and write the answer (page 88)

(i) $2 + $5

Step 1: 2 + 5 = 7.

Answer: $7

(ii) $12 + $13

Step 1: 12 + 13 = 25.

Answer: $25

(iii) 20 pounds + 30 pounds

Step 1: 20 + 30 = 50.

Answer: 50 pounds

Review Exercise — 3. Solve the following (page 88)

(i) Rs. 40 - Rs. 15

Step 1: 40 - 15 = 25.

Answer: Rs. 25

(ii) Riyal 17 - Riyal 5

Step 1: 17 - 5 = 12.

Answer: Riyal 12

(iii) Rs. 28 - Rs. 18

Step 1: 28 - 18 = 10.

Answer: Rs. 10

Review Exercise — 4. Fozia bought a story book for Rs. 45 and a play dough for Rs. 50. How much money did she spend altogether? (page 88)

Rs. 45 + Rs. 50

Step 1: Price of book = Rs. 45.

Step 2: Price of play dough = Rs. 50.

Step 3: Total money spent = 45 + 50 = 95.

Answer: Rs. 95

Unit 5 — Fractions (pages 89 to 99)

Students' learning outcomes (page 89): After completing this unit, students will be able to: identify the name and write unit fraction, non-unit fractions, like fractions and unlike fractions of a discrete set of objects using pictorial representation; compare and order unit fractions and like fractions with denominators up to 10 using >, < and = signs; add and subtract like fractions within 1 whole (for example, 1/4 + 2/4 = 3/4); know and recognize that tenths arise by dividing an object into 10 equal parts and in dividing single digit numbers and quantities by 10 (using concrete and pictorial representation).

Fractions from a cake (page 90)

A man brought a cake for the children. When he started cutting the cake, one child was observing keenly. He asked his father why he was cutting the cake into two pieces. His father explained to him that each piece of the cake is called a half, or 1/2. One whole = 1. Half = 1/2. One fourth = 1/4 (the book prints 'One forth'). Key point: how many parts of a whole — the top number is the numerator, which says how many parts are used; the bottom number is the denominator, which says how many parts the whole is divided into.

Fraction — Kinds of fractions (page 91)

A fraction is a part of a whole. A fraction can be written as 2/3 = numerator over denominator. (i) Unit fraction: those fractions which have '1' as numerator are called unit fractions, e.g. 1/2, 1/4, 1/6, shown with one part shaded. (ii) Non-unit fraction: those fractions which have a numerator other than one are called non-unit fractions, e.g. 2/3, 3/4, 5/7, shown with 2, 3 and 5 parts shaded. (iii) Like fraction: fractions with the same denominator are called like fractions, e.g. 1/6, 3/6, 4/6, shown on bars of 6 parts. NOTE — BOOK ERROR: the like-fraction definition is printed as 'which do not have same denominator'; the examples (all over 6) and the unlike-fraction definition on the next page show the intended word is 'do' — like fractions HAVE the same denominator.

Unlike fractions (page 92)

(iv) Unlike fraction: those fractions which do not have the same denominator are called unlike fractions, e.g. 2/6, 4/9, 1/4, shown on bars of 6, 9 and 4 parts.

Exercise 5.1 — 1. Tick the unit fractions and cross the non-unit fractions (page 92)

Fraction — unit or non-unit

(i) 1/3 — Unit — tick

(ii) 2/3 — Non-unit — cross

(iii) 4/6 — Non-unit — cross

(iv) 3/4 — Non-unit — cross

(v) 1/4 — Unit — tick

Exercise 5.1 — 2. Tick the like fractions and cross the unlike fractions (page 92)

Pair — like or unlike

(i) 2/5, 3/5 — Like — tick (same denominator 5)

(ii) 3/4, 2/6 — Unlike — cross (denominators 4 and 6)

(iii) 5/6, 4/8 — Unlike — cross (denominators 6 and 8)

(iv) 2/6, 3/6 — Like — tick (same denominator 6)

Comparing and ordering Fractions (page 93)

Comparing fractions: if two fractions have the same denominator, then the fraction with the greater numerator is greater. Example: Ajmal ate 3 pieces of pizza out of 5; Nasreen ate 2 pieces of pizza out of 5. Compare 3/5 and 2/5: denominator 5 is the same, but numerator 3 is greater than 2 (3 > 2). Therefore 3/5 is greater than 2/5, so 3/5 > 2/5. Hence Ajmal ate more pieces of pizza than Nasreen. We can also write 2/5 < 3/5 (2/5 is less than 3/5). Ordering fractions: ordering fractions means arranging fractions from the smaller to the bigger, or from the bigger to the smaller.

For fractions with the same denominator, the bigger fraction is the one having the greater numerator. Example: 3/7, 5/7, 2/7, 4/7 — select the fraction with the smallest numerator, i.e. 2/7, then the next, and so on. We get 2/7 < 3/7 < 4/7 < 5/7 (smaller to larger values); 5/7 > 4/7 > 3/7 > 2/7 (larger to smaller values); 5/7 = 5/7 = 5/7 = 5/7 (equal values, as printed).

Exercise 5.2 — 1. Compare fractions with <, > and = (page 94)

Pair — sign

(i) 1/2 ___ 3/2 — 1/2 < 3/2

(ii) 3/5 ___ 3/5 — 3/5 = 3/5

(iii) 2/3 ___ 2/3 — 2/3 = 2/3

(iv) 5/7 ___ 2/7 — 5/7 > 2/7

(v) 3/5 ___ 2/5 — 3/5 > 2/5

(vi) 6/9 ___ 6/9 — 6/9 = 6/9

(vii) 5/10 ___ 3/10 — 5/10 > 3/10

(viii) 2/3 ___ 1/3 — 2/3 > 1/3

(ix) 3/8 ___ 3/8 — 3/8 = 3/8

NOTE: Exercise 5.2 on this page is numbered 1, then 5, 6, 7 — items 2, 3 and 4 do not appear in the book at this point; nothing has been invented to fill them.

Exercise 5.2 — 5. Write the fraction of each figure and put <, > or = between them (page 94)

Q: (i) Left grid (2 of 12 squares shaded) and right grid (5 of 12 squares shaded):

A: 2/12 < 5/12.

Q: (ii) Left circle (top half shaded) and right circle (bottom half shaded):

A: 1/2 = 1/2.

Q: (iii) Left hexagon (3 of 4 parts shaded) and right hexagon (1 of 4 parts shaded):

A: 3/4 > 1/4.

Exercise 5.2 — 6. Arrange the following from larger to smaller (page 94)

Q: 4/7, 5/7, 1/7, 2/7, 6/7 from larger to smaller:

A: 6/7 > 5/7 > 4/7 > 2/7 > 1/7.

Exercise 5.2 — 7. Write from smaller to larger (page 94)

Q: 3/8, 4/8, 2/8, 5/8, 7/8 from smaller to larger:

A: 2/8 < 3/8 < 4/8 < 5/8 < 7/8.

Unit fractions up to 1/10 (page 95)

Fraction — number of parts — name

1/2 — 2 — One-half

1/3 — 3 — One-third

1/4 — 4 — One-fourth

1/5 — 5 — One-fifth

1/6 — 6 — One-sixth

1/7 — 7 — One-seventh

1/8 — 8 — One-eighth

1/9 — 9 — One-ninth

1/10 — 10 — One-tenth

Each row's shape shows one part shaded out of the stated number of parts.

Addition of like fractions (page 96)

To add fractions with the same denominator, we add the numerators. Example: 2/6 + 3/6 = 5/6, shown on a circle of 6 parts with 5 parts shaded.

Subtraction of like fractions (page 96)

To subtract fractions with the same denominators, we subtract the numerators. Example: 4/7 - 3/7 = 1/7, shown on bars of 7 parts: 4 shaded minus 3 shaded leaves 1 shaded.

Exercise 5.3 — 1. Add the following (page 96)

Sum — answer

(i) 3/7 + 2/7 — 5/7

(ii) 2/5 + 1/5 — 3/5

(iii) 5/8 + 2/8 — 7/8

(iv) 3/9 + 4/9 — 7/9

Exercise 5.3 — 2. Colour the figures according to the fraction (page 97)

Q: (i) 2/8 on a grid of 8 squares:

A: Colour 2 squares.

Q: (ii) 4/7 on a circle of 7 parts:

A: Colour 4 parts.

Q: (iii) 3/5 on a triangle of 5 parts:

A: Colour 3 parts.

Q: (iv) 3/4 on a square of 4 parts:

A: Colour 3 parts.

Exercise 5.3 — 3. Subtract the following (page 97)

Difference — answer

(i) 5/7 - 4/7 — 1/7

(ii) 3/5 - 2/5 — 1/5

(iii) 6/8 - 2/8 — 4/8

(iv) 6/9 - 4/9 — 2/9

I have Learnt (page 97)

Fraction shows parts of a whole; the fractions with 1 numerator are called unit fractions; the fractions with numerator other than 1 are called non-unit fractions; the fractions with same denominators are called like fractions; the fractions with different denominators are called unlike fractions; comparing and ordering like fractions and unit fractions; addition and subtraction of like fractions. Vocabulary used: Fraction, denominator, numerator, unit fraction, non-unit fraction, like fraction, unlike fraction. NOTE: the 'I have Learnt' box confirms the corrected like-fraction definition (same denominators), against the misprint on page 91.

Review Exercise — 1. Tick the correct option (page 98)

(i) Identify the unit fraction:

✓ 1/6

2/5

3/4

4/3

(ii) Identify the non-unit fraction:

✓ 3/9

1/5

1/7

6/7

(iii) The pair of like fractions:

✓ 2/7, 3/7

2/5, 4/8

2/4, 3/2

2/3, 3/4

(iv) 3/7 + 2/7 is equal to:

✓ 5/7

5/14

6/49

7/49

(v) Identify the smallest fraction (from 3/5, 5/5, 7/5, 2/5):

3/5

5/5

7/5

✓ 2/5

NOTE on Review 1 (ii): option (d) 6/7 is also a non-unit fraction; option (a) 3/9 is ticked as the first non-unit option. NOTE: Review Exercise numbering on this page runs 1, then 3, 4, 5 — item 2 does not appear; nothing has been invented to fill it.

Review Exercise — 3. Write fraction for the following figures (page 98)

Figure — fraction

(i) 2 by 2 grid, 3 squares shaded (given in the book) — 3/4

(ii) 3 by 3 grid, 4 squares shaded (top row 3, middle-left 1) — 4/9

(iii) 2 by 5 grid, 7 squares shaded (top row last 2, bottom row all 5) — 7/10

(iv) Circle of 8 parts, 2 parts shaded — 2/8

(v) Square of 8 triangles, 3 triangles shaded — 3/8

(vi) Bar of 8 cells, last cell shaded — 1/8

Review Exercise — 4. Solve the following (page 99)

Sum / difference — answer

(i) 3/8 + 7/8 — 10/8

(ii) 4/5 + 3/5 — 7/5

(iii) 7/12 + 9/12 — 16/12

(iv) 9/8 - 5/8 — 4/8

(v) 2/7 - 1/7 — 1/7

(vi) 8/15 - 3/15 — 5/15

NOTE: items (i), (ii) and (iii) give improper fractions above 1 whole, although the unit's teaching examples stay within 1 whole; the sums are given as computed, not reduced or capped.

Review Exercise — 5. Colour the figures according to the given fractions (page 99)

Q: (i) 5/8 on a bar of 8 cells:

A: Colour 5 cells.

Q: (ii) 9/12 on a grid of 12 squares:

A: Colour 9 squares.

Q: (iii) 1/8 on a grid of 8 squares:

A: Colour 1 square.

Q: (iv) 3/8 on a circle of 8 parts:

A: Colour 3 parts.

Unit 6 — Measurement: Length, Mass, Capacity

(pages 100 to 124)

Students' learning outcomes (page 100): After completing this unit, students will be able to: compare the lengths of different objects using standard unit of length (meter and centimeter) using <, > and = signs; recognize and use the standard units of length (metre and centimetre) to measure and record the length of different objects; add and subtract lengths given in same units to solve real world problems; recognize and use the standard units of mass (kilogram and gram) to measure and record the mass of different objects; compare the mass of different objects using standard units of mass (kilogram and gram) using <, > and = signs; add and subtract mass given in the same units to solve real world problems; compare the capacity of different objects using standard units (Liter and Milliliter) using <, > and = signs; recognize and use the standard units of capacity (liter and milliliter) to measure and record the capacity of different objects; add and subtract capacities given in the same units to solve real world problems; read and write temperature to

the nearest appropriate unit i.e. (degree C) using pictorial representations and relating temperature scale to number line; compare and order temperature using <, > and = signs.

Measurement of Length (page 101)

In our daily life we measure the length of various objects, like cloths, ropes, wires, height, etc. Metre (m): the standard unit of length is meter. The symbol 'm' is used for meter. 1 meter = 100 centimeter. Key point: usually meter is used to measure long length, like length and width of rooms, white boards, long tables, height of doors, height of trees and windows (pictured: a door, a table and a tree).

Centimetre (cm) (page 102)

Centimetre is a smaller unit of length; its symbol is 'cm'. Key point: usually centimetre is used for short length, pencils, pen, books, etc. The pictured pencil measured against the ruler is 15 cm long: the length of the pencil is 15 cm. Find the length of your book and eraser and write it down below the given pictures: this is the pupil's own measuring activity — measure your own book and your own eraser with a ruler and write the lengths in centimetres (no fixed answer is printed in the book).

Comparing the lengths of objects (page 103)

Let us measure the length of the given objects. Length of the comb is 9 cm. The length of the fish is 6 cm. Let us compare the length of the comb and the length of the fish: length of comb > length of fish; 9 cm > 6 cm. Length of comb is greater than length of fish. Now compare the length of the duster and pen: length of duster = 10 cm; length of pen = 10 cm; length of duster = length of pen; 10 cm = 10 cm. The lengths of duster

and pen are equal.

Comparing the length of a cell phone and a remote (page 104)

The length of a remote is 8 cm. The length of a cell phone is 6 cm. Let us compare the lengths of remote and cell phone: length of cell phone < length of remote; 6 cm < 8 cm. The length of cell phone is less than the length of remote.

Exercise 6.1 — 1. Write the suitable unit (Metre or Centimetre) in boxes to measure the following objects (page 105)

Object — suitable unit

(i) The height of class door — Metre

(ii) The length of your geometry box — Centimetre

(iii) Length of a van — Metre

(iv) Length of your arm — Centimetre

(v) Height of your desk — Centimetre

Exercise 6.1 — 2. Write the length of the given objects in the blanks (page 106)

Q: Length of sharpener (left edge at 0, right edge at the 1.5 cm mark):

A: 1.5 cm.

Q: Length of key (tip at 0, round end just past the 3 cm mark):

A: About 3 cm.

Q: Length of rope (from 0 to the 8 cm mark):

A: 8 cm.

Q: Length of pen (tip at 0, end at the 8 cm mark):

A: 8 cm.

Exercise 6.1 — 3. Read the length of the above objects and fill in the blanks using '>', '<' and = symbols (page 106)

Comparison — sign

(i) Length of sharpener ___ length of key — 1.5 cm < 3 cm

(ii) Length of rope ___ length of key — 8 cm > 3 cm

(iii) Length of rope ___ length of pen — 8 cm = 8 cm

Addition and subtraction of measurement of length (pages 106-107)

Example: Ahmad bought 13 m cloth from one shop and 15 m from another shop. How much meter of cloth has he bought altogether? Cloth bought from first shop = 13 m; cloth bought from other shop = + 15 m; total cloth = 28 m. Example: the length of the class room is 7 m and the length of the school hall is 18 metre. How much is the classroom shorter than the school hall? Length of school hall = 18 m; length of class = - 7 m; room is shorter by = 11 m.

Exercise 6.2 — 1. Solve the following (page 107)

Sum — answer

(i) 8 m + 7 m — 15 m

(ii) 14 m + 22 m — 36 m

(iii) 37 m + 42 m — 79 m

(iv) 60 m + 12 m — 72 m

Exercise 6.2 — 2. The length of red carpet is 12 m and brown carpet is 8 m. What is the total length of both red and brown carpets? (page 107)

12 m + 8 m

Step 1: Length of red carpet = 12 m.

Step 2: Length of brown carpet = 8 m.

Step 3: Total length of carpets = 12 + 8 = 20.

Answer: 20 m

Exercise 6.2 — 3. Samreen bought 35 cm long blue ribbon and 27 cm green ribbon. Find the total length of both ribbons (page 107)

35 cm + 27 cm

Step 1: Length of blue ribbon = 35 cm.

Step 2: Length of green ribbon = 27 cm.

Step 3: Total length of ribbons = 35 + 27 = 62.

Answer: 62 cm

Exercise 6.2 — 4. The length of Ahmad's garden is 85 m and Aslam's garden is 60 m. How much is Ahmad's garden greater than Aslam's garden? (page 108)

85 m - 60 m

Step 1: Length of Ahmad's garden = 85 m.

Step 2: Length of Aslam's garden = 60 m.

Step 3: Difference = 85 - 60 = 25.

Answer: 25 m greater

Measurement of mass (page 108)

To measure the accurate mass of objects, we use the standard unit of mass (weight measuring machine pictured). Key point: we use it to measure the mass of objects like vegetables, fruits, wheat, biscuits, etc. Kilogram (kg): kilogram is the standard unit of mass; its symbol is 'kg'. Pictured: a child on a scale reading 20 kg, and a wheat bag on a scale reading 5 kg.

Gram (g) (page 109)

Gram is a smaller unit of mass; its symbol is 'g'. Key point: we use gram to measure the mass of lighter objects like pencil, paper, medicine (tablets), gold, etc. 1 kg = 1000 g. Mass of lemon is 50 g; mass of apple is 100 g. Tick the object with greater mass: a mobile phone and a brick are pictured — the brick has the greater mass, so tick the brick.

Comparing masses (page 110)

Q: Football and balloon — tick the object with greater mass:

A: Football (a football is much heavier than a balloon).

Compare the mass of book and geometry box: the book reads 300 g and the geometry box reads 100 g. Book has greater mass than the geometry box: 300 g > 100 g. Rice bag reads 5 kg and flour bag reads 15 kg, so the flour bag has the greater mass: 15 kg > 5 kg.

Comparing masses — tea bag and bean bag (page 111)

Q: Tea bag reads 2 kg and bean bag reads 10 kg. The greater mass is:

A: Bean bag (10 kg).

Q: The smaller mass is:

A: Tea bag (2 kg).

Exercise 6.3 — 1. Identify which objects are measured in kilogram (kg) and which of them are measured in gram (g) (page 111)

Object — unit

Apple — Gram

Sharpener — Gram

Tea bag (1 kg pack pictured) — Kilogram

Toy car — Gram

Hen — Kilogram

Chocolate bar — Gram

Exercise 6.3 — 2. Read the mass of following objects and fill in the blanks (pages 111-112)

Object — mass

(i) Watermelon — 5 kg

(ii) Melon — 3 kg

(iii) Brick — 4 kg

Exercise 6.3 — 3. Compare the following using the signs <, > and = (page 112)

Comparison — sign

Melon (3 kg) ___ brick (4 kg) — 3 kg < 4 kg

Watermelon (5 kg) ___ melon (3 kg) — 5 kg > 3 kg

Addition and subtraction of masses (pages 112-113)

The weight of Ali is 22 kg and the weight of Haris is 26 kg. What is the

total mass? Weight of Ali = 22 kg; weight of Haris = + 26 kg; total weight of Ali and Haris = 48 kg. Bilal bought 25 kg mangoes and 15 kg apples. What is the difference between the mass of mangoes and apples? Mass of mangoes = 25 kg; mass of apples = 15 kg; difference in mass = 10 kg. The weight of Bilal is 56 kg and the weight of his friend is 38 kg. What is the total weight of both? Weight of Bilal = 56 kg; weight of friend = + 38 kg; total weight = 94 kg.

Exercise 6.4 — 1. Ahmad bought 80 g of red pepper and 45 g of black pepper. What is total mass of both peppers? (page 113)

80 g + 45 g

Step 1: Mass of red pepper = 80 g.

Step 2: Mass of black pepper = 45 g.

Step 3: Total mass = 80 + 45 = 125.

Answer: 125 g

NOTE: the book's box label for this item reads 'Total mass of paper'; the problem is about peppers, and 'peppers' is used here.

Exercise 6.4 — 2. A shopkeeper has 800 kg of rice. He sold 480 kg out of it. How much rice was left with him? (page 113)

800 kg - 480 kg

Step 1: Total weight of rice = 800 kg.

Step 2: Weight of sold rice = 480 kg.

Step 3: Weight of left rice = 800 - 480 = 320.

Answer: 320 kg left

Exercise 6.4 — 3. Weight of Taniya's bag is 850 g and mass of Saniya's bag is 775 g. Find the difference (page 113)

850 g - 775 g

Step 1: Weight of Taniya's bag = 850 g.

Step 2: Weight of Saniya's bag = 775 g.

Step 3: Difference = 850 - 775 = 75.

Answer: 75 g

Measurement of Capacity (page 114)

Capacity tells us how much liquid a container can hold. Tick the containers having more capacity: among the pictured jug, glass, cup, teapot and milk cartons, the jug is ticked in the book (it holds the most). To measure the capacity of containers, we use the standard unit of capacity. Liter (l): the standard unit of capacity is litre; its symbol is 'l'. Milliliter (ml): to measure a small quantity of liquid, we use milliliter; its symbol is 'ml'. 1 litre = 1000 millilitre; 1 l = 1000 ml.

Comparing Capacity (page 115)

Jug is larger than glass. It can hold 1 l of juice and the glass holds 250 ml of juice. Jug can hold more juice than the glass: 1 l > 250 ml. 250 ml juice is smaller than 500 ml juice; the 250 ml pack holds less juice than the 500 ml pack: 250 ml < 500 ml. 500 ml juice is equal to the 500 ml milk: 500 ml = 500 ml.

Exercise 6.5 — 1. Write suitable unit of capacity for the following objects (page 116)

Object — suitable unit

Water tank — Litre

Glass of juice — Millilitre

Bucket — Litre

Spoon — Millilitre

Petrol pump — Litre

Cup of tea — Millilitre

Exercise 6.5 — 2. Compare the following capacities using <, > and = (page 116)

Comparison — sign

(i) 12 l ketchup ___ 18 l water — 12 l < 18 l

(iii) 57 l petrol ___ 30 l oil — 57 l > 30 l

(iv) 1 l water ___ 750 ml drink — 1 l > 750 ml (1 l = 1000 ml)

NOTE: Exercise 6.5 item 2 is numbered (i), (iii), (iv) — item (ii) does not appear on the page; nothing has been invented to fill it.

Addition and subtraction of capacity (page 117)

A milkman bought 40 l milk from one dairy farm and 35 l from another dairy farm. How much milk did he buy? Milk bought from one dairy = 40 l; milk bought from other dairy = + 35 l; total milk bought = 75 l. There is 350 l petrol in a tank. If 180 l petrol is taken off, how much petrol is left in the tank? Total petrol in the tank = 350 l; petrol taken off = - 180 l; petrol left = 170 l.

Exercise 6.6 — 2. A dairy farm produces 160 l milk in the morning and 120 l milk in the evening. How much milk is produced in one

day? (page 117)

160 l + 120 l

Step 1: Milk produced in the morning = 160 l.

Step 2: Milk produced in the evening = 120 l.

Step 3: Milk produced in one day = 160 + 120 = 280.

Answer: 280 l

Exercise 6.6 — 3. There was 950 l water in a tanker, 430 l water was consumed in watering flowers. How much water left in the tanker? (page 117)

950 l - 430 l

Step 1: Water in tanker = 950 l.

Step 2: Water consumed = 430 l.

Step 3: Water left = 950 - 430 = 520.

Answer: 520 l

NOTE: Exercise 6.6 on this page starts at item 2 — item 1 does not appear; nothing has been invented to fill it.

Exercise 6.6 — 4. A bottle contains 250 ml fever syrup. Out of 250 ml syrup, 130 ml syrup is given to children. How much syrup left in the bottle? (page 118)

250 ml - 130 ml

Step 1: Total syrup = 250 ml.

Step 2: Syrup given to children = 130 ml.

Step 3: Syrup left = 250 - 130 = 120.

Answer: 120 ml

Temperature (page 118)

Temperature is the measure of hotness and coldness. The unit of temperature is called degree centigrade, written as degree C. Two kinds of thermometer are pictured: a common thermometer and a digital thermometer (reading 37.6 degrees C). Normal body temperature of a human body is 37 degrees C (teacher note, page 120).

Number line — Temperature of different cities (page 119)

Observe the temperature on the thermometer and mark it on the number line. Sibi: 40 degrees C; Islamabad: 15 degrees C; Quetta: 5 degrees C. NOTE — PRINT INCONSISTENCY: the Sibi thermometer's red column rises to about the 55 mark on its scale, while its printed label says 40 degrees C; the printed labels are used here (Islamabad and Quetta columns match their labels). Comparing and ordering temperature: Akbar wants to know the difference of temperature between Ziarat and Sibi. Ziarat reads 20 degrees C and Sibi reads 30 degrees C on the pictured thermometers: 20 degrees C < 30 degrees C. The temperature is 10 degrees C higher in Sibi than in Ziarat.

Activity and example (page 120)

Activity: measure the temperature of your classroom in the morning and afternoon and compare both temperatures — this is the pupil's own measuring activity (no fixed answer is printed). Example: Jameel measured the temperature of his room. Temperature in morning = 15 degrees C; temperature in afternoon = 30 degrees C; temperature at night = 25 degrees C. Ordered: 15 degrees C < 25 degrees C < 30 degrees C.

Exercise 7 — 1. Write and identify the temperature on a number line (page 120)

Q: (i) The horizontal thermometer reads 20 degrees C and the number line arrow points at 20:

A: 20 degrees C.

Exercise 7 — 1 (continued). Write and identify the temperature on a number line (page 121)

Q: (ii) The horizontal thermometer's blue column (bulb on the right, scale running 100 down to 0 left to right) reaches the 35 mark:

A: 35 degrees C — mark 35 on the number line.

Q: (iii) The column reaches the 15 mark:

A: 15 degrees C — mark 15 on the number line.

Exercise 7 — 2. Observe the temperature and compare their values (page 121)

Q: Left thermometer reads about 35 degrees C and right thermometer reads about 20 degrees C:

A: 35 degrees C > 20 degrees C.

I have Learnt (page 122)

Measuring the length of objects by using standard units of length; meter (m) and centimeter (cm) are standard units for length; meter (m) is used to measure the long length while centimeter (cm) is used for short lengths; addition and subtraction of the lengths in same unit; measuring the mass of objects by using the standard units of mass; kilogram (kg) and gram (g) are standard units of mass; addition and subtraction of

mass in same units; to measure capacity we use standard units of capacity; liter (l) and milliliter (ml) are standard units of capacity; addition and subtraction of capacity in same units; recognizing how to measure the temperature of coldness and hotness. Vocabulary used: Length, height, meter, centimeter, mass, weights, kilogram, gram, capacity, liter, milliliter, temperature, Celsius degree, thermometer, hot, cold.

Review Exercise — 1. Choose the correct answer (pages 122-123)

(i) The length of your pen is measured in:

- Metre
- ✓ Centimetre
- Litre
- Gram

(ii) 1 metre is equal to:

- 1 cm
- 10 cm
- ✓ 100 cm
- 1000 cm

(iii) Standard unit of mass is:

- ✓ kg
- g
- mg
- km

(iv) 27 degrees C ___ 19 degrees C:

- ✓ >
- <
- =

x

(v) Litre is the standard unit of:

Mass

Length

Temperature

✓ Capacity

(vi) We buy petrol in standard unit:

kg

ml

✓ l

cm

(vii) Mass of rice is measured in:

ml

mg

✓ kg

g

Review Exercise — 2. Circle the suitable unit to measure the objects (page 123)

Object — suitable unit

(i) Length of your lunch box — cm

(ii) Mass of gold — g

(iii) Mass of sharpener — g

(iv) Capacity of milk — l

(v) Height of tree — m

NOTE: Review Exercise numbering on this page runs 1, 2, 4, 5, 6, 6 —

item 3 does not appear, and item 6 appears twice (rice problem and Quetta/Gawadar problem); both are solved below in order.

Review Exercise — 4. There are two water bottles of same capacity, one water bottle can hold 6 liters of water. How many liters of water can both bottles hold? (page 123)

6 l + 6 l

Step 1: Capacity of first bottle = 6 l.

Step 2: Capacity of second bottle = 6 l (same capacity).

Step 3: Total capacity = 6 + 6 = 12.

Answer: 12 l

Review Exercise — 5. The length of red electric cable is 172 m while length of yellow cable is 85 m. Which wire is longer and by how much? (page 123)

172 m - 85 m

Step 1: Length of red cable = 172 m.

Step 2: Length of the yellow cable = 85 m.

Step 3: Red cable is longer by 172 - 85 = 87.

Answer: Red cable is longer by 87 m

Review Exercise — 6. Ahmad sold 185 kg rice in the morning and 65 kg in the evening. How much rice did he sell altogether? (page 124)

185 kg + 65 kg

Step 1: Rice sold in the morning = 185 kg.

Step 2: Rice sold in the evening = 65 kg.

Step 3: Total rice sold = 185 + 65 = 250.

Answer: 250 kg

Review Exercise — 6 (second item 6). The temperature of Quetta is 25 degrees C. Temperature of Gawadar is 40 degrees C. Compare the temperature of both cities (page 124)

Q: Quetta 25 degrees C and Gawadar 40 degrees C:

A: 25 degrees C < 40 degrees C. Gawadar is warmer than Quetta by 40 - 25 = 15 degrees C.

Unit 7 — Time (pages 125 to 135)

Students' learning outcomes (page 125): After completing this unit, students will be able to: read and write time in hours and minutes (with five minutes interval, half past, quarter past, and quarter two) from analogue and digital clock; show time in hours and minutes on an analogue clock; recognize intervals of time (for instance to estimate a rough calculation of the time taken by a particular event or task); use Solar and Islamic calendar to find the particular day, date in real life situation.

Hours and Minutes (page 126)

The dial of the clock is divided into 12 big equal parts. One big part represents an hour. The digital clock pictured shows 12:45 and the analogue clock pictured shows the hour hand near 11 with the minute hand at 12. Key point: 1 hour = 60 minutes; 1 day = 24 hours. Reading

time on analogue and digital clocks: 12:00 is twelve o'clock (both hands at 12); 12:15 is quarter past twelve (hour hand at 12, minute hand at 3).

Reading time continued (page 127)

12:30 is half past twelve (hour hand between 12 and 1, minute hand at 6); 12:45 is quarter to one (hour hand near 1, minute hand at 9). Activity 1 — clock reading with 5 minutes interval: observe the digital and analogue clock and compare their interval. 12:25: hour hand at 12, minute hand at 5 (5 times 5 = 25 minutes). 12:30: hour hand at 12, minute hand at 6 (30 minutes). 12:35: hour hand at 12, minute hand at 7 (35 minutes). NOTE: in this activity the pink hand is the hour hand and the black hand is the minute hand, which is how the analogue clocks match the digital times.

Activity 2 — Look at the clocks and write the time (page 128)

We got up early in the morning: 06:00 am (given in the book; pink hour hand at 6, black minute hand at 12). Children go to school: 08:00 am (pink hour hand at 8, black minute hand at 12). Children come back from school: 02:00 am as printed in the box (pink hour hand at 2, black minute hand at 12). We take our dinner: 08:00 am as printed in the box (pink hour hand at 8, black minute hand at 12). NOTE — BOOK QUIRK: all four boxes are pre-printed 'am', although coming back from school and dinner are afternoon/evening activities; the times are given here exactly as the boxes print them. Key point: a.m. (ante meridiem) runs from midnight to 12:00 noon; p.m. (post meridiem) runs from 12:00 noon to midnight.

Exercise 7.1 — 1. Write the time from digital clock (page 129)

Q: Left analogue clock (short hour hand at 9, long minute hand at 6; the

thin hand is the seconds hand and is ignored):

A: 9:30.

Q: Right analogue clock (short hour hand at 12, long minute hand at 3):

A: 12:15.

NOTE: the question says 'from digital clock' but the clocks pictured are analogue, with blank digital boxes below them; the times above are the readings to write in those boxes.

Exercise 7.1 — 2. Show the given time on analogue clock (page 129)

Q: 12:05:

A: Short hour hand just past 12, long minute hand at 1 (5 minutes).

Q: 12:10:

A: Short hour hand just past 12, long minute hand at 2 (10 minutes).

Q: 12:15:

A: Short hour hand at 12, long minute hand at 3 (15 minutes, quarter past twelve).

Exercise 7.1 — 3. Read and write the time (page 130)

Clock — time

Pink clock (short hand between 4 and 5, long hand at 6) — 4:30

Blue clock (short hand at 1, long hand at 3) — 1:15

Lavender clock (short hand at 12, long hand at 6) — 12:30

Blue clock (short hand between 8 and 9, long hand at 9) — 8:45

The thin third hand on these clocks is the seconds hand and is ignored for the hour-and-minute readings.

Activity — intervals of time (page 131)

Estimate and fill in from your own experience (no fixed answers are printed in the book): (i) I eat an apple in ___ minutes. (ii) I can run across the school playground in ___ minutes. (iii) I brush my teeth for ___ minutes. (iv) I remain in school for ___ hours.

Solar Calendar 2025 (page 131)

The solar calendar for 2025 is printed month by month (January to December). Key point: some months have 30 days and some months have 31 days. Activity: how many Fridays come in December? Identify their dates. From the December 2025 calendar (1 December is a Monday): the Fridays are 5, 12, 19 and 26 December — 4 Fridays in December.

Islamic Calendar (page 132)

There are 12 months and 354 or 355 days in a lunar year. It is also known as the Hijri and Islamic calendar. The printed calendar maps each solar month of the year to its Hijri months (for example January = Rajab/Sha'ban, March = Ramadan/Shawwal, June = Dhu'l-Hijjah/Muharram, July = Muharram/Safar). Each cell shows the solar date in large print and the Hijri date in small print beneath it.

I have Learnt (page 133)

Showing of time in hours and minutes in an analogue clock; recognizing intervals of time; recognizing different months of solar year; recognizing different months of Islamic calendar. Vocabulary used: Analogue clock, digital clock, hour, minutes, solar calendar, lunar calendar, months, days, clockwise.

Review Exercise — 1. Write the time below each clock (page 133)

Clock — time

First clock (short hand at 9, long hand at 12) — 9:00

Second clock (short hand at 3, long hand at 6) — 3:30

Third clock (short hand at 12, long hand at 3) — 12:15

Review Exercise — 2. Draw the given time on the clocks (page 133)

Q: 5:30:

A: Short hour hand between 5 and 6, long minute hand at 6.

Q: 12:00:

A: Both hands at 12.

Q: 01:15:

A: Short hour hand at 1, long minute hand at 3.

Review Exercise — 3. Read and write the time (page 134)

Clock — time

First clock (short hand at 7, long hand at 6) — 7:30

Second clock (short hand between 10 and 11, long hand at 9) — 10:45

Third clock (short hand at 9, long hand at 3) — 9:15

NOTE — HEADING MISMATCH: this item is headed 'Differentiate between Solar calendar and Islamic calendar' but the items pictured are clocks, so they are solved here as times. The calendar difference is stated in the unit text: the solar year has 12 months with 30 or 31 days, while the Islamic (lunar, Hijri) year has 12 months and 354 or 355 days.

Review Exercise — 4. Use the calendar to answer the questions (June 2025, page 134)

Q: (i) How many Fridays are in the month?

A: 4 Fridays: 6, 13, 20 and 27 June.

Q: (ii) What is the date on the second Monday?

A: 9 June (Mondays are 2 and 9 June; the second is 9 June).

Q: (iii) What day is the last day of the June?

A: Monday, 30 June.

Q: (iv) What day is on the 25th of June?

A: Wednesday.

Review Exercise — 5. Fill the boxes with the help of calendar (page 135)

The printed table is laid out unclearly, so its given cells are reproduced here and the determinable answers are given from the 2025 solar calendar on page 131. Month August: August 2025 begins on Friday 1 August. Date 14th in January: 14 January 2025 is a Tuesday (January begins on Wednesday 1 January). Date 31st that is a Friday and the month's last Friday: in 2025 this fits January (Friday 31 January) and October (Friday 31 October). Your birthday: the pupil's own date — find its day from the calendar and write it in. NOTE: the table's row labels do not state which single question each row asks, so the answers above give every reading the printed cells support rather than guessing one intended row.

Review Exercise — 6. Find and write the days of 6th September and 25th December (page 135)

Q: 6th September 2025 (September begins on Monday 1 September):

A: Saturday.

Q: 25th December 2025 (December begins on Monday 1 December):

A: Thursday.

Review Exercise — 7. Write the name of days (Lunar Calendar) (page 135)

Read from the Islamic calendar on page 132 (solar date in large print, Hijri date in small print): 7th Safar = Saturday (Safar 1 falls on Sunday 27 July, so Safar 7 falls on Saturday 2 August); 1st Muharram = Friday (Muharram 1 falls on Friday 27 June; the July panel shows 10 Muharram boxed on Sunday 6 July); 19th Rajab = Sunday (Rajab 1 falls on Wednesday 1 January, so Rajab 19 falls on Sunday 19 January); 22nd Shawwal = Monday (Shawwal 1 falls on Monday 31 March, so Shawwal 22 falls on Monday 21 April).

Review Exercise — 8. Write the name of months in which we celebrate both Eids (page 135)

Q: The two Eid months in the Islamic calendar:

A: Shawwal (Eid-ul-Fitr) and Dhu'l-Hijjah (Eid-ul-Adha).

Unit 8 — Geometry (pages 136 to 151)

Students' learning outcomes (page 136): After completing this unit, students will be able to: recognize, identify and draw 2D shapes (rectangle, square, circle, triangle, semicircle, quarter circle) and patterns with respect to their characteristics; identify of perpendicular and parallel

lines; recognize and identify 3D shapes in different orientations; make 3D shapes using varied modeling materials (cube, cuboid, cone, cylinder, sphere) with respect to their characteristics; describe the position, direction and movement of an object including moving clockwise, anticlockwise, quarter, half and three quarter using appropriate positional language (inside, outside, above, below, over, under, near, far, before, after, left, right and in front of/behind etc.); recognize turn as a rotation; identify and differentiate between a straight and curved lines; identify horizontal and vertical lines; identify quarter turns.

2-D Shapes (page 137)

Shape — characteristics

Rectangle — It has two equal opposite sides and four corners

Square — It has four equal sides and four corners

Circle — It has round shape with no sides and no corners

Triangle — It has three sides and three corners

Semi-circle — It is half of the circle

Quarter circle — It is one-fourth of a circle (printed 'one-forth of a cirlce')

Activity 1 and Activity 2 (page 138)

Activity 1: ask students to find and list objects around them that match different shapes. Examples given: Circle — clock, bangle, plate, etc. Square — tile, window pane, etc. Rectangle — book, door, matchbox, etc. Triangle — some pictures, slice of sandwich, etc. Activity 2: give students cut-outs of circles, squares, rectangles and triangles; ask them to make pictures using shapes like a house, flowers and animals, etc. Both activities are the pupil's own work — the lists and pictures depend on the

objects the pupil finds.

Patterns (page 139)

A pattern is something that repeats in a special order, like colors, shapes. Let us learn different patterns on a square grid: pattern in different directions (pink triangles pointing in different directions); pattern in different sizes (blue triangles of different sizes); pattern in different shapes (pink triangles and rectangles alternating); pattern in different size and color (blue triangles with a small yellow triangle).

3D Shapes — Cube and Cuboid (page 140)

Cube (pictured as a dice, with vertex, edge and face labelled, and a dotted 'Trace' outline): the cube has 6 square faces, 12 edges with same length and 8 vertices. Cuboid (pictured as a gift box, with edge, vertex and face labelled, and a dotted 'Trace' outline): the cuboid has 6 rectangular faces, 12 edges and 8 vertices.

3D Shapes — Cone, Cylinder and Sphere (page 141)

Shape — edges, faces, vertices

Cone (birthday cap, vertex, face and edge labelled) — 1 edge, 2 faces, 1 vertex

Cylinder (measuring jug, edge and face labelled) — 2 edges, 3 faces, 0 vertex

Sphere (red ball, face labelled) — 0 edge, 1 face, 0 vertex

Exercise 8.1 — 1. Match the following (page 142)

Shape pictured — name

Yellow square — Square

Blue circle — Circle

Pink rectangle — Rectangle

Pink triangle — Triangle

Exercise 8.1 — 2. Write down the sides and corners of the given shapes (page 142)

Shape — sides — corners

Square — 4 — 4 (given in the book)

Rectangle — 4 — 4

Circle — 0 — 0

Triangle — 3 — 3

Exercise 8.1 — 3. Write any two names of 3D shapes (page 142)

Q: Any two names from this unit are correct, for example:

A: Cube and sphere (cuboid, cone and cylinder are also correct).

Exercise 8.1 — 4. Write vertex, edge and face of cube (page 143)

Q: The cube (pictured as a dice):

A: 8 vertices, 12 edges, 6 square faces.

Exercise 8.1 — 5. Write and identify the name and numbers of vertices, edges and faces of cuboid (page 143)

Q: The cuboid (pictured as a matchbox):

A: 8 vertices, 12 edges, 6 rectangular faces.

Exercise 8.1 — 6. Sphere has ___ edge, ___ face and ___ vertex (page 143)

Q: Sphere:

A: 0 edge, 1 face, 0 vertex.

Exercise 8.1 — 7. Complete the following patterns according to shapes and color (page 143)

This is a drawing task on the square grid: the started pattern combines pink triangles (pointing down and up), a pink rectangle column, blue squares and yellow squares. Continue the same shapes in the same colours and order across the empty part of the grid. No single written answer is printed in the book for this item.

The turning of an object (page 144)

(i) Clockwise direction: the direction of movement of a body along the direction of hands of clock is called a clockwise direction. (ii) Anti-clockwise direction: the direction of movement of a body against the direction of the hands of clock is called anti-clockwise direction. (iii) Quarter turn: the turn of a body around a quarter either clockwise or anti clockwise is called a quarter turn. (iv) Half turn: the turn of a body around half circle either clockwise or anti clockwise is called a half turn.

Turns continued and Position and direction (page 145)

(v) Three quarter turn: the turn of a body around 3 quarters either clockwise or anti clockwise is called a three quarter turn. (vi) Full turn: the turn of a body around the circle either clockwise or anti clockwise is called a full turn. Position and direction — (i) Inside and outside: the girl sitting in the box is inside; the girl standing beside the box is outside.

Position and direction continued (pages 146-147)

(ii) Below and above: the girl sitting under the table is below; the girl sitting on the table is above. (iii) Far and near: the old man and woman sitting apart on the bench are far; sitting close together they are near. (iv) Left and right: the arrows point left and right. (v) Behind and in front of: the boy standing behind the box is behind the box; the boy standing before the box is in front of the box.

Lines (page 147)

(i) Curved lines: a curved line is a line that bends and changes direction as it goes (pictured as a wavy line and a winding road). (ii) Straight lines: a straight line is a path that goes in one direction without bending or curving (pictured as a straight horizontal line, a straight vertical line, and a straight road).

Perpendicular and Parallel Lines (page 148)

(iii) Perpendicular lines: a horizontal line and a vertical line intersect at a point labeled 'O'. These lines are called perpendicular lines (pictured as a crossroads). (iv) Parallel lines: two lines that never meet (intersect) are known as parallel lines. A railway track is made of two parallel lines.

Exercise 8.2 — 1. Match the following (page 148)

Label — diagram it matches

Perpendicular Lines — The crossing horizontal and vertical lines (first diagram)

Parallel Lines — The two horizontal lines that never meet (second diagram)

Straight Lines — The two straight slanted lines (fourth diagram)

Curve Lines — The wavy bending line (third diagram)

NOTE: the diagrams on the right are not in the same order as the labels on the left — the curve diagram comes third and the straight-lines diagram fourth; the matches above follow the diagrams, not the order.

Exercise 8.2 — 2. Tick the right one (page 149)

(a) The turn of a body from point A to point B is (shaded quarter of the circle):

✓ Quarter turn

Three quarter turn

(b) The turn of a body from point C to point D is called (shaded three quarters of the circle):

Half turn

✓ Three quarter turn

I have Learnt (page 149)

Recognizing the 2-D shapes (rectangle, square, circle, triangle, semi-circle and quarter-circle) with respect to their characteristics; identifying perpendicular and parallel lines; differentiating between straight and curved lines; identifying horizontal and vertical lines; identifying the position, direction and movement of object; recognizing and identifying 3-D shapes (cube, cuboid, cone, cylinder, sphere) with respect to their characteristics. Vocabulary used: 2-D shapes, 3-D shapes, square, rectangle, circle, semi-circle, triangle, cube, cuboid, cylinder, sphere, cone, parallel line, horizontal lines, vertical lines, perpendicular lines, turn, quarter, half, three quarter, patterns.

Review Exercise — 1. Complete the following table by writing the name and number of sides and corners (page 150)

Shape pictured — name — corners — sides

Yellow square — Square — 4 — 4

Pink rectangle — Rectangle — 4 — 4

Pink triangle — Triangle — 3 — 3

Blue circle — Circle — 0 — 0

Review Exercise — 2. Complete the Chart (page 150)

Shape pictured — name — faces — edges — vertices

Dice — Cube — 6 — 12 — 8

Gift box — Cuboid — 6 — 12 — 8

Birthday cap — Cone — 2 — 1 — 1

Measuring jug — Cylinder — 3 — 2 — 0

Ball — Sphere — 1 — 0 — 0

NOTE: the chart's middle column is headed 'No. of Corners'; the counts above use the unit's own edge counts (cube 12, cuboid 12, cone 1, cylinder 2, sphere 0) for that column, since the unit teaches faces, edges and vertices. Read literally as corners, that column would equal the vertices column (cube 8, cuboid 8, cone 1, cylinder 0, sphere 0).

Review Exercise — 3. Write the total numbers of straight and curved lines (page 151)

Q: First figure (a slanted shape with straight top and bottom and curved sides):

A: Number of straight lines = 2; number of curved lines = 2.

Q: Second figure (a dome on a straight baseline):

A: Number of straight lines = 1 (the baseline); number of curved lines = 1 (the arc).

Review Exercise — 4. Write 'y' for parallel lines and 'z' for perpendicular lines in the following pairs (page 151)

Pair — label

(i) Two horizontal lines that never meet — y (parallel)

(ii) Two diagonal lines crossing as an X — Neither — the lines cross but not at a right angle, so they are neither parallel (y) nor perpendicular (z) under the book's definitions; the book's two-label scheme has no fitting label for this pair

(iii) Two slanted lines that never meet — y (parallel)

(iv) A horizontal line crossed by a vertical line — z (perpendicular)

Unit 9 — Information Handling (pages 152 to 160)

Students' learning outcomes (page 152): After completing this unit, students will be able to: read and interpret data using pictograph, bar graphs and tally charts and represent data using tally charts (including real world problem); describe the likelihood that everyday events will occur using mathematical language (i.e. impossible, less likely, more likely, unlikely, certain).

Pictograph (page 153)

A pictograph uses pictures or symbols to represent data. A teacher

conducted a survey and collected data on students' favorite fruit. How can we represent that data in the form of graph? There are different ways to represent data on graph, i.e. pictograph, bar graph and tally chart. Example: students' favorite fruits can be represented as follows. Each symbol stands for 2 objects (as printed), with a star = student. The table shows: Orange — 2 stars; Strawberry — 3 stars; Melon — 1 star; Apple — 4 stars.

Pictograph questions (page 153)

Q: How many students like apple? (filled in the book)

A: 4.

Q: How many students like orange more than melon?

A: Orange 2 minus melon 1 = 1 more student.

Q: How many less students like melon than apple?

A: Apple 4 minus melon 1 = 3 less students.

Q: What is the total number of students?

A: Orange 2 + strawberry 3 + melon 1 + apple 4 = 10 students.

NOTE — BOOK CONTRADICTION: the pictograph header says each symbol stands for 2 objects, but the book's own filled answer counts 4 stars as 4 students for apple. The answers above follow the book's filled answer (1 star = 1 student). If the '2 objects' rule were applied instead, the counts would double (apple 8, total 20).

Bar Graph (page 154)

A bar graph is a graph that uses vertical or horizontal bars to represent the numerical data. Example: there are 40 students in a class. They play Cricket, Hockey, Football and Volleyball games; we can represent the

number of students who play different games by using a bar chart. The bars read: Cricket 15, Hockey 5, Football 10, Volleyball 10 (which add to the stated 40 students).

Bar graph questions (page 154)

Q: How many students play cricket? (filled in the book)

A: 15.

Q: Most favourite game in a graph?

A: Cricket (15 students, the tallest bar).

Q: Less favourite game in a graph?

A: Hockey (5 students, the shortest bar).

Q: How many students play Volleyball?

A: 10.

Tally Chart (page 155)

A tally chart is a table that represents data by using tally marks. It helps to record data quickly and easily. One mark stands for 1 student, two marks stand for 2 students, and similarly a group of five marks stands for 5 students. Example: favorite subjects of students by tally chart. English: three groups of five plus two marks = 17. Math: two groups of five plus two marks = 12. Science: four marks = 4. Urdu: one group of five plus four marks = 9.

Tally chart questions (page 155)

Q: What is the total number of students? (filled in the book)

A: 42 (English 17 + math 12 + science 4 + Urdu 9 = 42).

Q: Which subject is the most favourite?

A: English (17 students).

Q: What is the total number of students who liked Urdu and English?

A: Urdu 9 + English 17 = 26 students.

Q: Which subject is liked by least number of students?

A: Science (4 students).

Probability (page 156)

Probability describes the chance of occurrence of an event: more likely, less likely, impossible, etc. Example: a student has a geometry box with 3 red color pencils, 2 blue color pencils and 1 yellow color pencil. If the student picks one pencil without looking, the chance of getting a red pencil is called the probability of picking red. In the same way, we can find the probability of picking a blue or yellow pencil too! There are different words to describe probability, on a scale from impossible, to less likely, to likely, to more likely, to certain. The book's bullets: to choose a pencil is certain; to choose a blue pencil is more likely; to choose a yellow pencil is less likely; to choose a white pencil is impossible. Key point: if an event is sure to happen, then it has a certain probability; if an event is sure not to happen, then it has impossible probability.

NOTE — BOOK INCONSISTENCY: the box holds 3 red, 2 blue and 1 yellow pencils, so red is actually the most likely single colour and blue is less likely than red; the book's bullet 'to choose a blue pencil is more likely' is reproduced as printed but should be read against the counts (red most likely, then blue, then yellow least likely). A box of blue and yellow pencils is also pictured at the foot of the page.

I have Learnt (page 157)

Interpreting pictograph, bar graph, and tally chart; describing likelihood of everyday events; identifying certain probability and impossible probability; differentiating between unlike, more like and less like.
Vocabulary used: Pictograph, bar graph, tally chart, likelihood, data, certain.

Review Exercise — 1. Given below is the data of favourite fruit of students. Draw a pictograph that represents the given information (page 157)

Data: Apple 12 students, Banana 20 students, Orange 2 students, Mango 13 students. Draw a pictograph with one symbol = one student: Apple 12 symbols, Banana 20 symbols, Orange 2 symbols, Mango 13 symbols. (If one symbol = 2 students were used instead, Mango's 13 would need a half symbol, which is why the one-symbol-per-student scale is the clean choice here.)

Review Exercise — 2. Describe probability of landing the needle on each colour; write impossible, more likely, less likely and certain events (page 157)

Needle landing — likelihood

When land on black (black covers half the spinner) — Likely (50% on the scale on page 158)

When land on any colour — Certain (the needle must land on some colour)

When land on all colours — Impossible (the needle cannot land on all colours at once)

When land on green or white (two small sections together) — Less likely (together they cover less than half the spinner)

Review Exercise — 3. The given tally chart represents the number of glasses of water students drink daily. Read and answer the questions (page 158)

Tally counts: Sheerin 4 glasses; Benazir 7 glasses (one group of five plus two); Hafsa 5 glasses (one group of five); Mahrung 9 glasses (one group of five plus four); Malala 11 glasses (two groups of five plus one).

Q: How many glasses of water did Mahrung drink?

A: 9 glasses.

Q: How many glasses of water did Malala and Hafsa drink altogether?

A: Malala 11 + Hafsa 5 = 16 glasses.

Q: How many more glasses of water did Benazir drink than Sheerin?

A: Benazir 7 - Sheerin 4 = 3 more glasses.

Review Exercise — 4. Choose the correct likelihood (certain, more likely, less likely and impossible) (page 158)

The scale printed on the page: 0% = impossible, 25% = less likely, 50% = likely, 75% = more likely, 100% = certain.

Statement — likelihood

The chance of the sunrise in the morning is ___ — Certain

It is ___ to sit outside in the month of June — Less likely (June heat makes sitting outside less likely)

It is ___ to wear jackets in winter — More likely

It is ___ to rain without clouds — Impossible

NOTE: the June and jacket items are judgement items with no fixed answer printed in the book; the answers above are the natural readings of the likelihood scale.

Review Exercise — 5. A survey conducted in the school asked the students to vote their favourite cricket team in Asia (page 159)

Bar chart readings: Pakistan 30, Afghanistan 15, Sri Lanka 20, Bangladesh 10, India 5.

Q: How many students like Pakistan team?

A: 30 students.

Q: How many students voted for Bangladesh team?

A: 10 students.

Q: The least likely team?

A: India (5 votes, the shortest bar).

Q: How many students liked Afghanistan?

A: 15 students.

Final page of the book (page 160)

Page 160 is headed 'Your valuable suggestions' and is a blank lined page for the reader's own suggestions. It contains no exercises, so there is nothing to solve on it. This completes the book: all units 1 to 9 and all exercises, pages 1 to 160, are covered in these notes.